

Chapter 6

Static Magnetic Fields



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6.1 Fundamental Postulates of Magnetostatics in Free Space

- ❖ The two fundamental postulates of magnetostatics are

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \qquad \nabla \cdot \mathbf{B} = 0$$

- where μ_0 is the permeability of free space:

$$\mu_0 = 4\pi \times 10^{-7} \quad (\text{H/m})$$

- And, $\mathbf{J}$ is the current density. Since the divergence of the curl of any vector field is zero, we obtain

$$\nabla \cdot (\nabla \times \mathbf{B}) \equiv 0 \quad \longrightarrow \quad \nabla \cdot \mathbf{J} = 0$$

- which is consistent with the equation of continuity.



• Equations for Steady Magnetic Fields in Free Space (integral form)

❖ Ampere's circuital law

circulation of the magnetic flux density in free space around any closed path is equal to μ_0 times the total current flowing through the surface bounded by the path.

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} \quad \longrightarrow \quad \oint_l \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

where μ_0 is the permeability of vacuum, and I is the current enclosed by the closed curve.

$$\nabla \cdot \mathbf{B} = 0 \quad \longrightarrow \quad \oint_S \mathbf{B} \cdot d\mathbf{S} = 0$$

1. The total magnetic flux through a **closed** surface is equal to **zero**.
2. The magnetic field lines are **closed everywhere**, with no beginning or end. This may be called the principle of **magnetic flux continuity**.

• Divergence and curl of steady magnetic field

Divergence of constant magnetic field (Differential form)

$$\nabla \cdot \vec{B}(\vec{r}) = 0$$

Curl of constant magnetic field (Differential form)

$$\nabla \times \vec{B}(\vec{r}) = \mu_0 \vec{J}(\vec{r})$$

Principle of continuity of magnetic flux (Integral form)

$$\oint_S \vec{B}(\vec{r}) \cdot d\vec{S} = 0$$

Ampere circuital law (Integral form)

$$\oint_C \vec{B}(\vec{r}) \cdot d\vec{l} = \mu_0 I$$

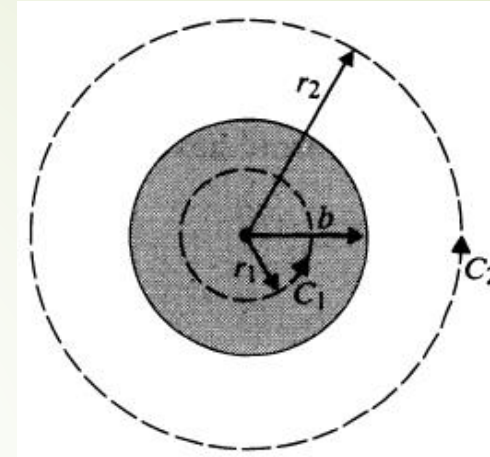
Conclusion:

- A steady magnetic field is a solenoidal and rotational field, which is a non-conservative field
- Current is the vortex source of the magnetic field
- The magnetic flux line is a closed curve with no start and end point

Example

Ex: An infinitely long, straight conductor with a circular cross section of radius b carries a steady current I . Determine the magnetic flux density both inside and outside the conductor.

Solution :



a) Inside the conductor:

$$\mathbf{B}_1 = \mathbf{a}_\phi B_{\phi 1}, \quad d\ell = \mathbf{a}_\phi r_1 d\phi$$
$$\oint_{C_1} \mathbf{B}_1 \cdot d\ell = \int_0^{2\pi} B_{\phi 1} r_1 d\phi = 2\pi r_1 B_{\phi 1}.$$

The current through the area enclosed by C_1 is

$$I_1 = \frac{\pi r_1^2}{\pi b^2} I = \left(\frac{r_1}{b}\right)^2 I.$$

Therefore, from Ampère's circuital law,

$$\mathbf{B}_1 = \mathbf{a}_\phi B_{\phi 1} = \mathbf{a}_\phi \frac{\mu_0 r_1 I}{2\pi b^2}, \quad r_1 \leq b. \quad (6-11a)$$



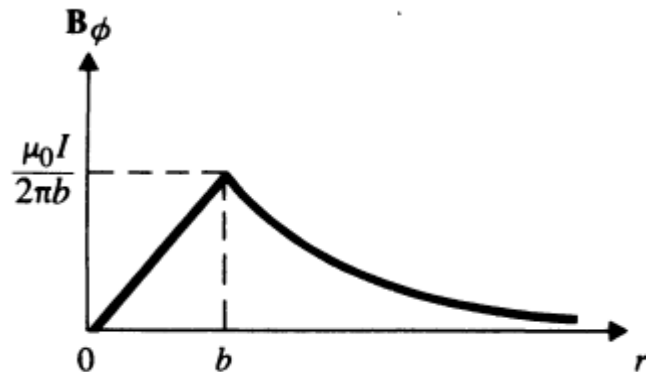
b) Outside the conductor:

$$\mathbf{B}_2 = \mathbf{a}_\phi B_{\phi 2}, \quad d\ell = \mathbf{a}_\phi r_2 d\phi$$
$$\oint_{C_2} \mathbf{B}_2 \cdot d\ell = 2\pi r_2 B_{\phi 2}.$$

Path C_2 outside the conductor encloses the total current I . Hence

$$\mathbf{B}_2 = \mathbf{a}_\phi B_{\phi 2} = \mathbf{a}_\phi \frac{\mu_0 I}{2\pi r_2}, \quad r_2 \geq b. \quad (6-11b)$$

Examination of Eqs. (6-11a) and (6-11b) reveals that the magnitude of $\mathbf{B}$ increases linearly with r_1 from 0 until $r_1 = b$, after which it decreases inversely with r_2 . The variation of B_ϕ versus r is sketched in Fig. 6-2(b).



Magnetic flux density of an infinitely long circular conductor carrying a current I

If it is not a solid cylindrical conductor but a very thin circular tube carrying a surface current density J_s , then it is obvious:

$$B = \begin{cases} 0, & r < b, \\ \mathbf{a}_\phi \frac{\mu_0 b}{r} J_s, & r > b. \end{cases}$$



6.2 Vector Magnetic Potential

- ➡ The divergence-free postulate of $\mathbf{B}$, assures that $\mathbf{B}$ is solenoidal.

$$\nabla \cdot \mathbf{B} = 0$$

- ❖ As a consequence, $\mathbf{B}$ can be expressed as the curl of another vector field, say $\mathbf{A}$, such that

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (T) \quad \longleftarrow \quad \nabla \cdot (\nabla \times \mathbf{A}) \equiv 0$$

$\mathbf{A}$: *vector magnetic potential*

- ❖ The definition of a vector field requires the specification of both its curl and its divergence.
- ❖ How do we choose divergence of $\mathbf{A}$?

The divergence of the magnetic vector potential can be chosen arbitrarily!

1. Definition of vector magnetic potential

$$\nabla \cdot \vec{A} = 0 \quad (\text{steady magnetic field})$$



Coulomb gauge

2. Differential equations of magnetic vector potential (general problems)

$$\left. \begin{aligned} \vec{B} &= \nabla \times \vec{A} \\ \nabla \times \vec{B} &= \mu_0 \vec{J} \end{aligned} \right\} \Rightarrow \nabla \times \nabla \times \vec{A} = \mu_0 \vec{J} \Rightarrow \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A} = \mu_0 \vec{J}$$

$\nabla \cdot \vec{A} = 0$

$$\nabla^2 \vec{A} = -\mu_0 \vec{J} \quad \text{--- Vector Poisson equation}$$

In the source free area :

$$\vec{J} = 0 \quad \Rightarrow \quad \nabla^2 \vec{A} = 0 \quad \text{--- Vector Laplace's Equation}$$

3. The problems in infinite uniform medium space (special problems)

➤ The analogy method to solve

$$\begin{aligned}\mathbf{A} &= A_1 \mathbf{e}_x + A_2 \mathbf{e}_y + A_3 \mathbf{e}_z \\ \mathbf{J} &= J_1 \mathbf{e}_x + J_2 \mathbf{e}_y + J_3 \mathbf{e}_z\end{aligned}$$

$$\nabla^2 \varphi(r) = -\frac{1}{\varepsilon} \rho(r)$$

$$\nabla^2 \mathbf{A} = -\mu_0 \mathbf{J}$$

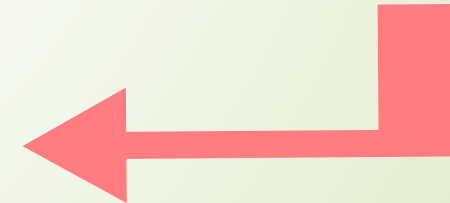


$$\begin{cases} \nabla^2 A_1 = -\mu_0 J_1 \\ \nabla^2 A_2 = -\mu_0 J_2 \\ \nabla^2 A_3 = -\mu_0 J_3 \end{cases}$$

$$\varphi(r) = \frac{1}{4\pi\varepsilon} \int_{V'} \frac{\rho(r') dv'}{R} + C$$

$$A_i(r) = \frac{\mu_0}{4\pi} \int_{V'} \frac{J_i(r') dv'}{R} + C_i$$

$$\mathbf{A}(r) = \frac{\mu_0}{4\pi} \int_{V'} \frac{\mathbf{J}(r') dv'}{R} + \mathbf{C}$$



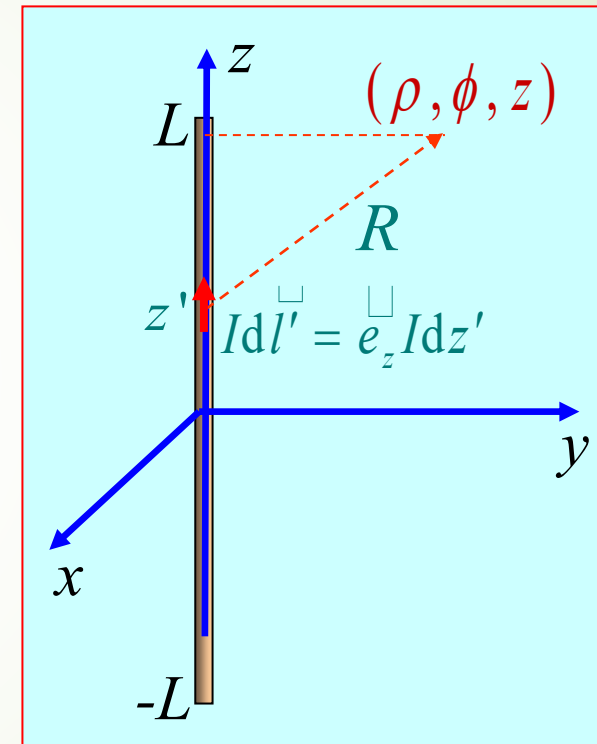
Ex. Find the magnetic vector potential of an infinite long line carrying current I long z axis.

Solution: the magnetic vector potential for $2L$ line, current element $Idl' = \vec{e}_z Idz'$

$$\begin{aligned} \text{to } P(\rho, \phi, z), \text{ the } R &= \sqrt{\rho^2 + (z - z')^2} \\ \vec{A}(\vec{r}') &= \vec{e}_z \frac{\mu_0 I}{4\pi} \int_{-L}^L \frac{1}{\sqrt{\rho^2 + (z - z')^2}} dz' \\ &= \vec{e}_z \frac{\mu_0 I}{4\pi} \ln[z' - z + \sqrt{\rho^2 + (z - z')^2}] \Big|_{-L}^L \\ &= \vec{e}_z \frac{\mu_0 I}{4\pi} \ln \frac{\sqrt{\rho^2 + (z - L)^2} - (z - L)}{\sqrt{\rho^2 + (z + L)^2} - (z + L)} \end{aligned}$$

The magnetic vector potential for $L \rightarrow \infty$

$$\vec{A}(\vec{r}) = \vec{e}_z \frac{\mu_0 I}{2\pi} \ln \frac{2L}{\rho} + C$$



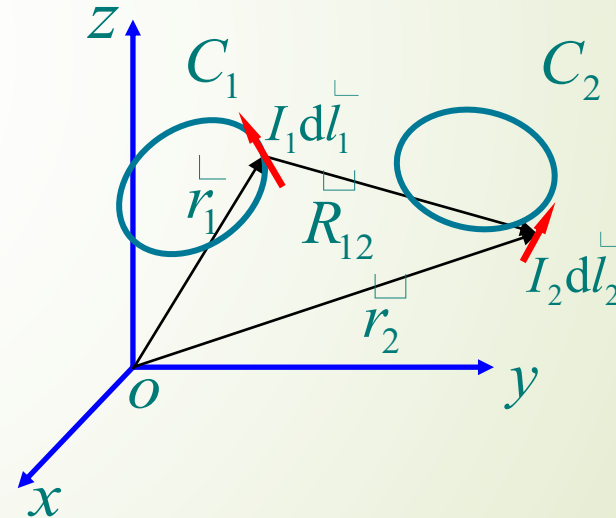
6.3 The Biot-Savart Law and Applications

In static magnetic field: The field has the effect on the current element located in a magnetic field.

1. Ampere's force law

$$\vec{F}_{12} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{I_2 d\vec{l}_2 \times (I_1 d\vec{l}_1 \times \vec{R}_{12})}{R_{12}^3}$$

Practice: $\vec{F}_{21} = -\vec{F}_{12}$



Discussion

- Ampère's force law obeys Newton's third law (the law of action and reaction).
- Ampère's Law describes the existence of forces between current-carrying circuits, but it does not specify how these forces are exerted.

2. Magnetic flux density $\vec{B}$

According to Ampere's force law

*Biot-Savart
Law*

$$\vec{F}_{12} = \oint_{C_2} I_2 d\vec{l}_2 \times \left(\frac{\mu_0}{4\pi} \oint_{C_1} \frac{I_1 d\vec{l}_1 \times \vec{R}_{12}}{R_{12}^3} \right) = \oint_{C_2} I_2 d\vec{l}_2 \times \vec{B}_1(\vec{r}_2)$$

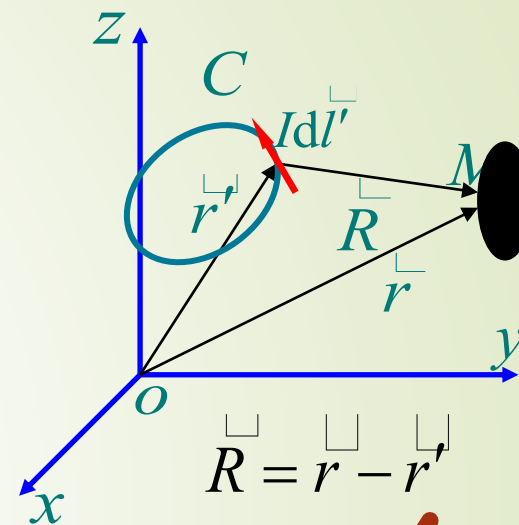
which: $\vec{B}_1(\vec{r}_2) = \frac{\mu_0}{4\pi} \oint_{C_1} \frac{I_1 d\vec{l}_1 \times \vec{R}_{12}}{R_{12}^3}$ is independent of I_2 , we call it magnetic flux density generated by I_1 , and its unit is T.

- *Biot-Savart Law describes how the magnetic force are exerted*
——by magnetic field

Magnetic flux density of different current models

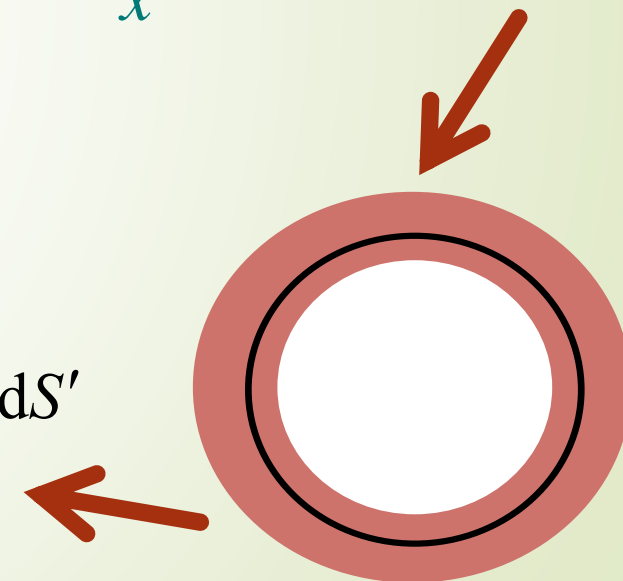
🏠 Magnetic flux density produced by arbitrary current loop C

$$dB(\vec{r}) = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \vec{R}}{R^3} \longrightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \oint_C \frac{Id\vec{l}' \times \vec{R}}{R^3}$$



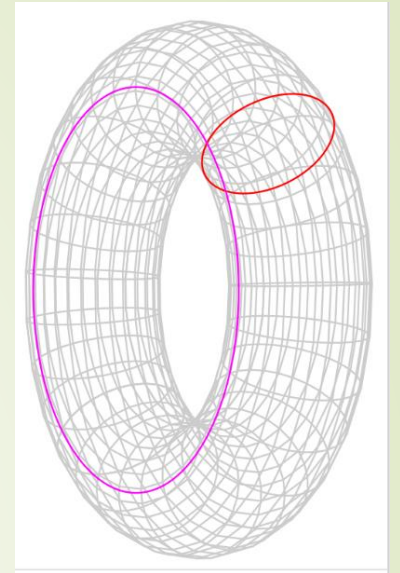
🏠 Magnetic flux density produced by surface current

$$dB(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{J}_s ds \times \vec{R}}{R^3} \longrightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_S \frac{\vec{J}_s(\vec{r}') \times \vec{R}}{R^3} dS'$$



Magnetic flux density produced by volume current

$$dB(\vec{r}) = \frac{\mu_0}{4\pi} \frac{J dV \times \vec{R}}{R^3} \longrightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{J(\vec{r}') \times \vec{R}}{R^3} dV'$$



Magnetic flux density produced by convection current

$$dB(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\rho \vec{v} \times \vec{R}}{R^3} \longrightarrow \vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \int_V \frac{\rho \vec{v} \times \vec{R}}{R^3} dV'$$

3. Magnetic flux densities of several typical current distributions

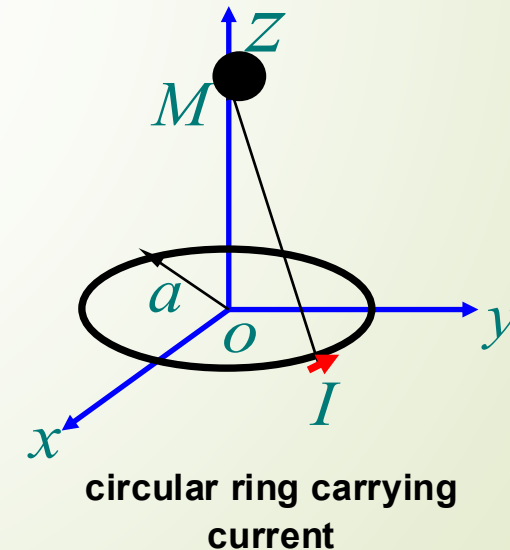
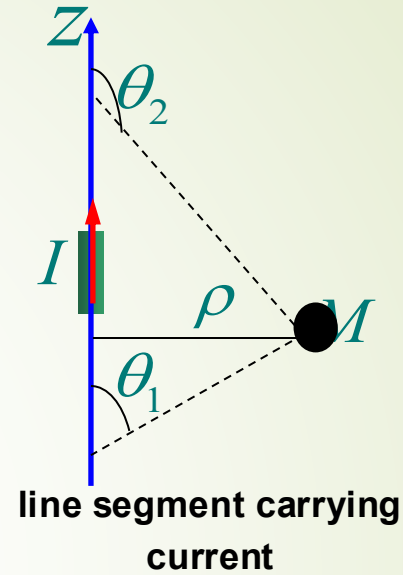
- The magnetic flux density of current carrying line segment :

$$\vec{B} = \vec{e}_\phi \frac{\mu_0 I}{4\pi\rho} (\cos \theta_1 - \cos \theta_2) \quad (\text{Finite long})$$

$$\vec{B} = \vec{e}_\phi \frac{\mu_0 I}{2\pi\rho} \quad (\text{Infinite long})$$

- Magnetic flux density on the axis of current carrying circular ring :

$$B(0,0,z) = \vec{e}_z \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}}$$



Ex. Calculate the magnetic flux density at arbitrary point at the center of the circular ring.

Solution : Establish a best coordinate system ,as shown in Figure .

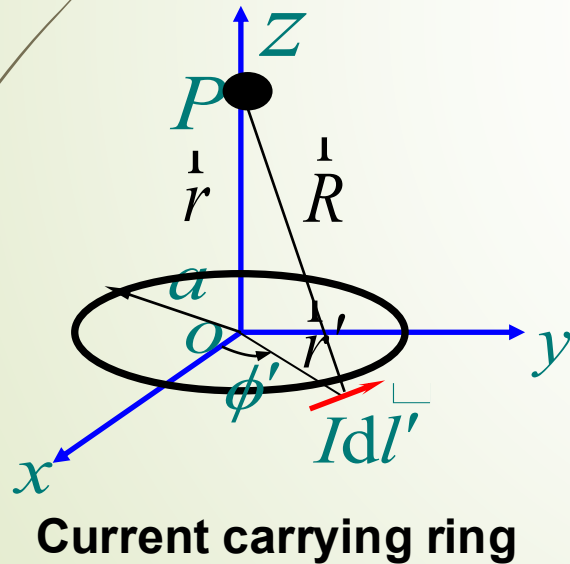
For the magnetic flux density at arbitrary point on the axis $P(0, 0, z)$, because:

$$Idl' = e_\phi I a d\phi' \quad r' = e_\rho a \quad r = e_z z$$

so

$$r - r' = e_z z - e_\rho a, \quad |r - r'| = (z^2 + a^2)^{1/2}$$

$$\begin{aligned} Idl' \times (r - r') &= e_\phi I a d\phi' \times (e_z z - e_\rho a) \\ &= e_\rho I a z d\phi' + e_z I a^2 d\phi' \end{aligned}$$



So: Magnetic flux density at $P(0, 0, z)$ is

$$B(z) = \frac{\mu_0 I a}{4\pi} \int_0^{2\pi} \frac{e_\rho z + e_z a}{(z^2 + a^2)^{3/2}} d\phi'$$

for $\int_0^{2\pi} e_\rho d\phi' = \int_0^{2\pi} (e_x \cos\phi' + e_y \sin\phi') d\phi' = 0$, so

$$B(z) = \frac{\mu_0 I a}{4\pi} \int_0^{2\pi} \frac{e_z a}{(z^2 + a^2)^{3/2}} d\phi' = e_z \frac{\mu_0 I a^2}{2(z^2 + a^2)^{3/2}}$$

Discussion

- Why the magnetic field at P only has Z component ?
- Where is the maximum magnetic flux density ?

At the center of the ring , as $z = 0$: $B(0) = e_z \frac{\mu_0 I}{2a}$

- What about the far field? (When $z \gg a$)

$$(z^2 + a^2)^{3/2} \approx z^3 \quad B = e_z \frac{\mu_0 I a^2}{2z^3}$$

Laws and methods of steady magnetic field to the methods of analyzing problems

Laws

- Current sources

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{R}}{R^3}$$

- Vortex source of the magnetic field

$$\nabla \times \vec{B}(\vec{r}) = \mu_0 \vec{J}(\vec{r})$$

- The circulation of the magnetic field is proportional to the current

$$\oint_C \vec{B}(\vec{r}) \cdot d\vec{l} = \mu_0 I$$

Methods

- Known current distribution, solve the distribution of magnetic field

Superposition principle

- Known the distribution of magnetic field

Conduct differential operation

- Known current distribution

Solving differential equations

- Known magnetic field distribution, solve its circulation

Solving integral equations

- Known current, solve its magnetic field

Solving integral equations

- A simple way to solve magnetic flux density
--using **Ampere's circuital law**

Condition: Problem has symmetry, the integral equation can be transformed to algebraic equation !

Example: When surface current density is $\vec{J}_s = \hat{e}_z J_{s0}$, find magnetic flux intensity produced by infinite current sheet.

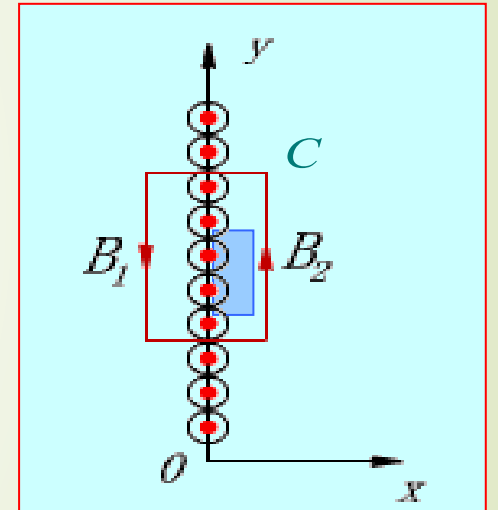
Solution: Establish a best coordinate system

so : $\vec{B}(r) = \vec{B}(x)$

Based on the symmetry, the loop integral is:

$$\oint_C \vec{B} \cdot d\vec{l} = B_1 l + B_2 l = \mu_0 J_{s0} l \quad \Rightarrow \quad \vec{B} = \begin{cases} \hat{e}_y \frac{\mu_0 J_{s0}}{2} & x > 0 \\ -\hat{e}_y \frac{\mu_0 J_{s0}}{2} & x < 0 \end{cases}$$

$$B_1 = B_2 = B$$



Ex. Find the magnetic flux density of the infinite long current carrying coaxial cable.

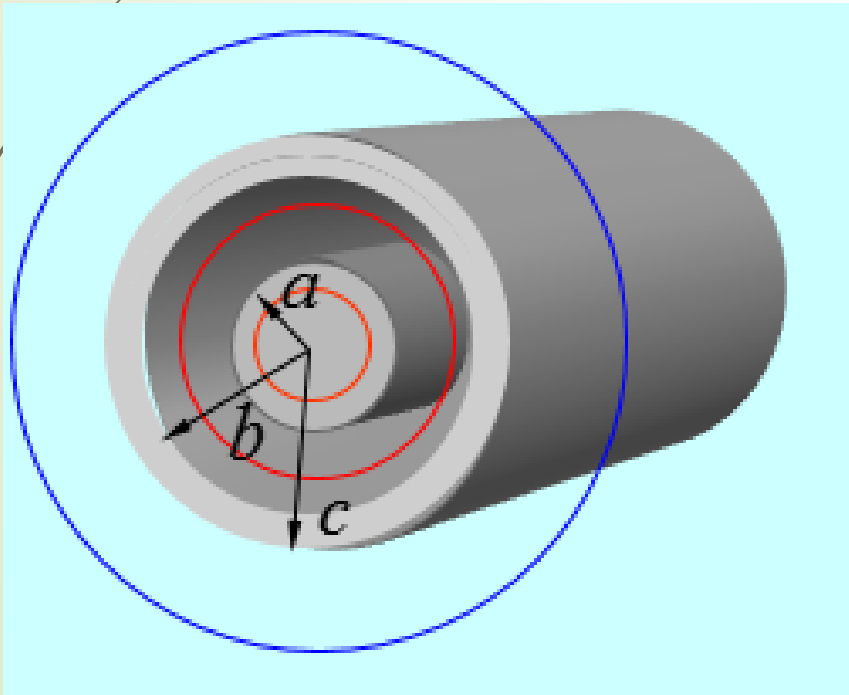
Solution: Choose cylindrical coordinate system

, so $\vec{B}(r) = e_\phi B(\rho)$

(1) $0 \leq \rho < a$

Choose ampere loop ($\rho < a$), the current of the cross link is

$$I_1 = \frac{I}{\pi a^2} \cdot \pi \rho^2 = I \frac{\rho^2}{a^2}$$



Applying **Ampere's circuital law**

$$2\pi\rho B_1 = \mu_0 \frac{I\rho^2}{a^2}$$



$$\vec{B}_1 = e_\phi \frac{\mu_0 I \rho}{2\pi a^2}$$

(2) $a \leq \rho < b$

$$2\pi\rho B_2 = \mu_0 I \quad \Rightarrow \quad B_2 = e_\phi \frac{\mu_0 I}{2\pi\rho}$$

(3) $b \leq \rho < c$

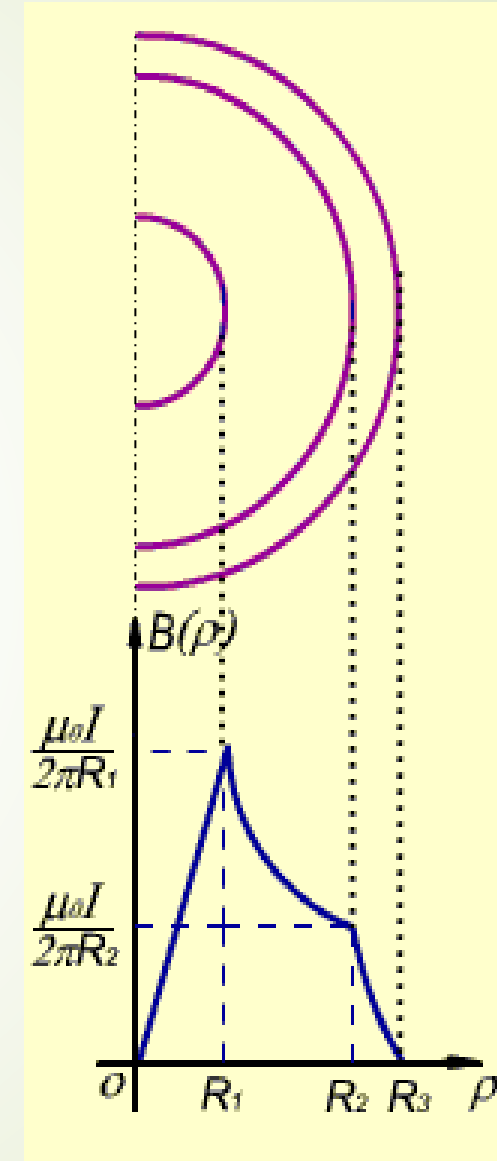
$$I_3 = I - I \frac{\rho^2 - b^2}{c^2 - b^2} = I \frac{c^2 - \rho^2}{c^2 - b^2}$$

Applying **Ampere's circuital law**

$$2\pi\rho B_3 = \frac{\mu_0 I (c^2 - \rho^2)}{c^2 - b^2} \quad \Rightarrow \quad B_3 = e_\phi \frac{\mu_0 I}{2\pi\rho} \cdot \frac{c^2 - \rho^2}{c^2 - b^2}$$

(4) $c \leq \rho < \infty$

$$I_4 = 0 \quad \Rightarrow \quad B_4 = 0$$



6.4 Magnetization and Equivalent Current Densities

1. Magnetization of medium

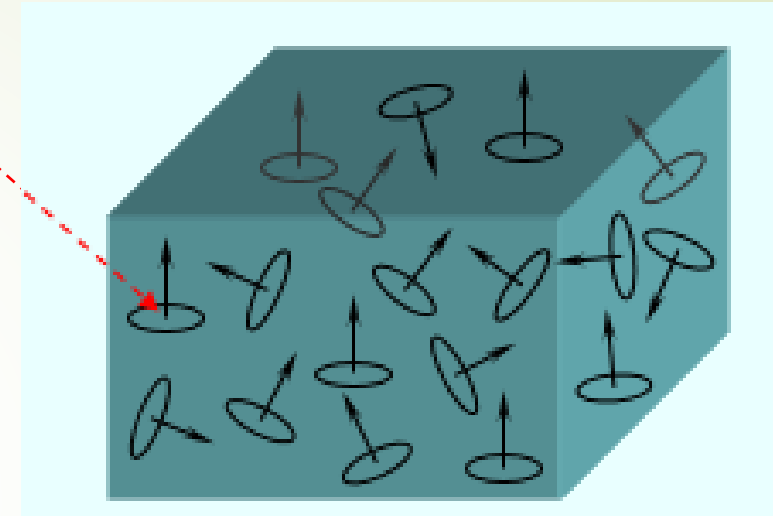
Conclusion

- 1) Under the effect of external magnetic field, the magnetic moments of the medium molecules are oriented, resulting in the magnetic phenomenon (showing magnetism).
- 2) The magnitude of the magnetization is determined by the number of magnetic moment in the medium.

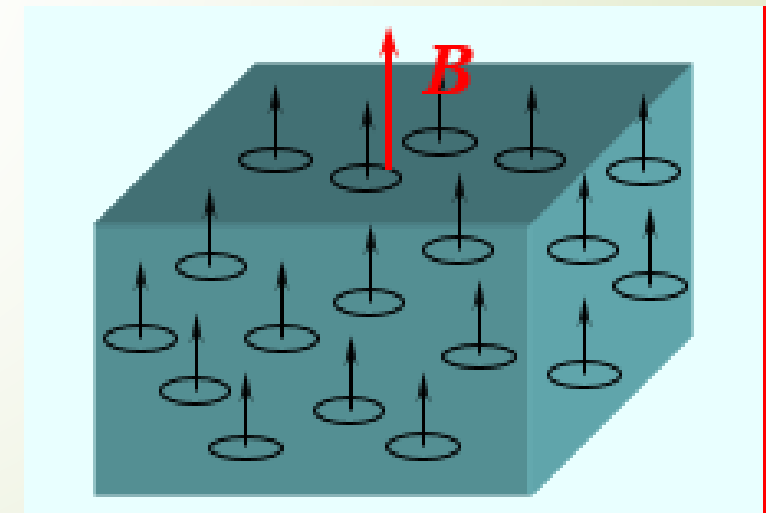
Definition of magnetic moment

$$\vec{p}_e = q_e \vec{l} \quad \vec{p}_m = i \Delta \vec{S}$$

$$\vec{p}_m = i \Delta \vec{S}$$



No external magnetic field

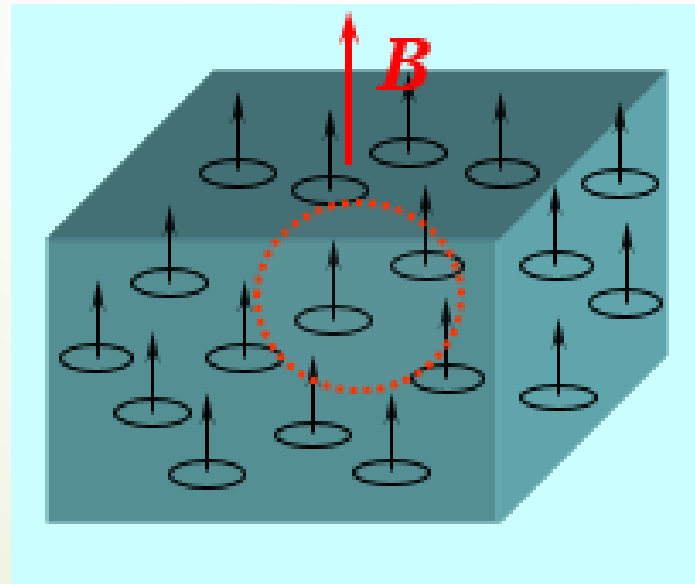


External magnetic field applied

2. Magnetization vector

Definition: Sum of the magnetic dipole moment per unit volume of the medium, that

$$\vec{M} = \lim_{\Delta v \rightarrow 0} \frac{\sum_i \vec{p}_{mi}}{\Delta v} \quad (A / m)$$

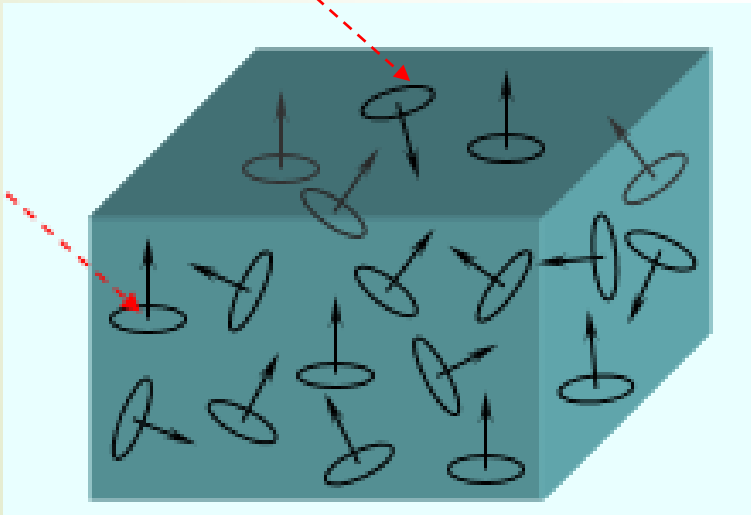


Discussion

1) When the medium has no external magnetic field

$$\sum_i \vec{p}_{mi} = 0 \Rightarrow \vec{M} = 0$$

$$\vec{p}_m = i\Delta S$$



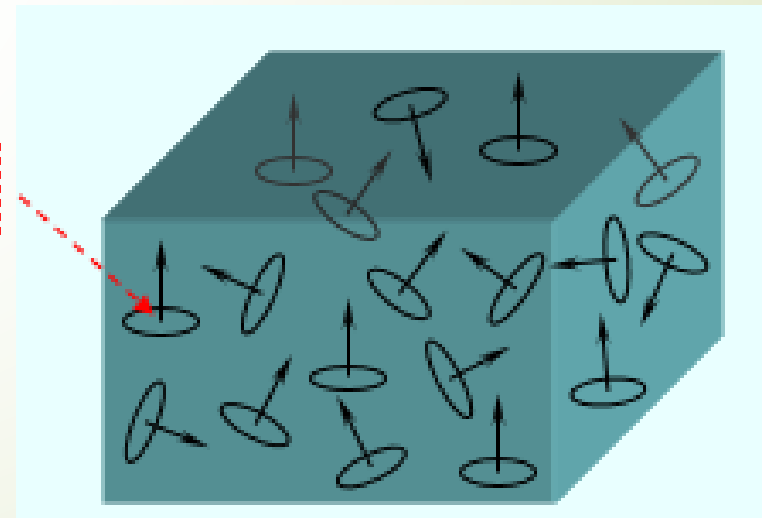
No external magnetic field

2) With external magnetic field

$$\begin{aligned} \vec{p}_{m1} &= \vec{p}_{m2} = \dots = \vec{p}_m \\ \Rightarrow \sum_i \vec{p}_{mi} &= N\vec{p}_m \\ \Rightarrow \vec{M} &= n\vec{p}_m \end{aligned}$$

There, n is the number of magnetized molecules in the unit volume

$$\vec{M} = n\vec{p}_m$$



No external magnetic field

Conclusion

- 1) The magnitude of the magnetization vector is related to the dielectric material
- 2) The magnitude of the magnetization is also related to the intensity of the applied field B_e
 - After the medium is magnetized, extra magnetic field B_m will be generated in the space
 - The total magnetic field B in the medium inside and outside space is

$$B = B_e + B_m$$

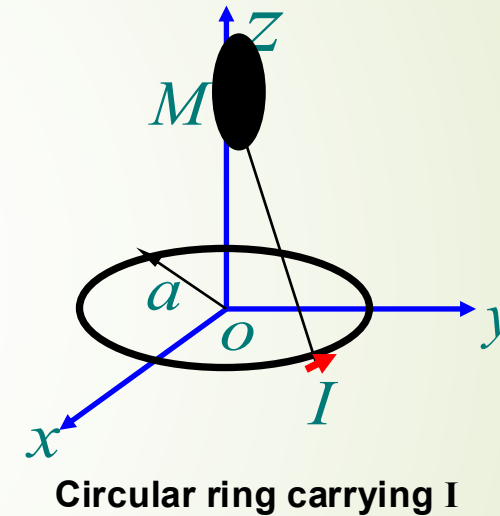
Experimental results:

For linear and isotropic media, M is proportional to H , that $M = \chi_m H$

where, $\chi_m (= 0; < 0; > 0)$ is called as the magnetic susceptibility of the medium

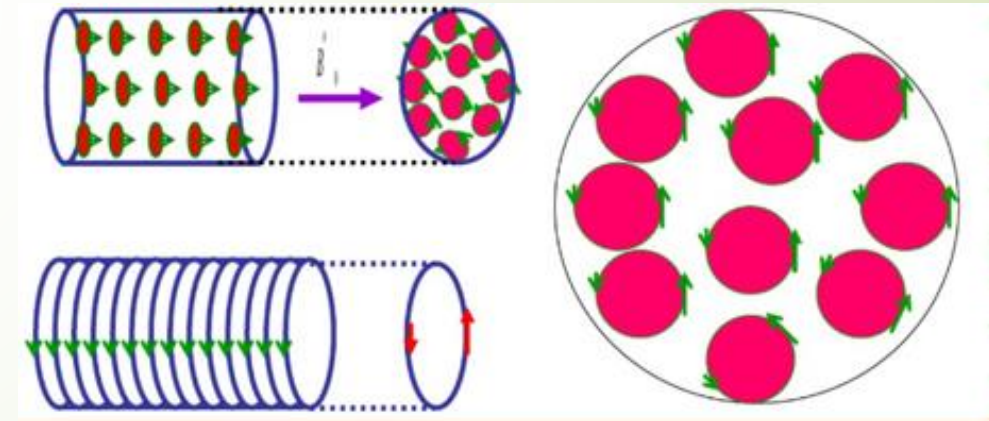
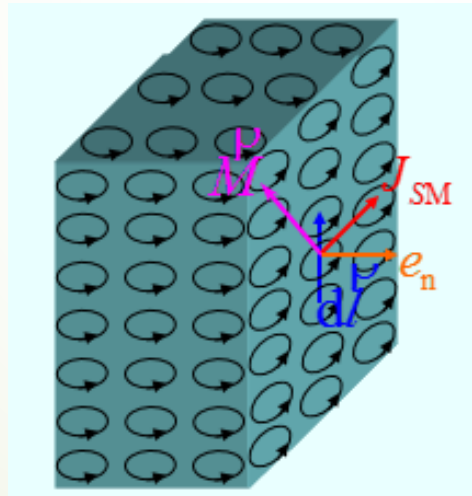
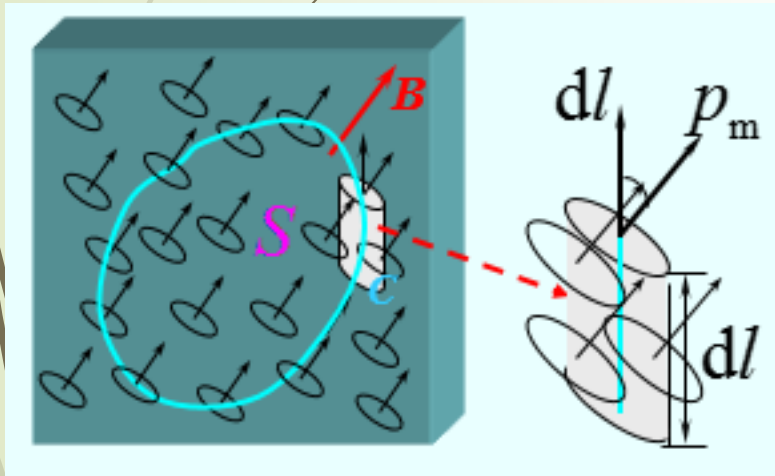
- Magnetic flux density on the axis of the current carrying circular ring :

$$\begin{aligned} B(0,0,z) &= e_z \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}} \\ &= \frac{\mu_0 p_m}{2\pi(a^2 + z^2)^{3/2}} \end{aligned}$$



Further discussion on magnetization phenomenon

- After medium is magnetized, net current distribution $\vec{J}_M$ could appear **internally**, that is the generation of volume magnetization current.
- After medium is magnetized, net current distribution $\vec{J}_{SM}$ could appear **at interface of medium**, that is the generation of surface magnetization current.



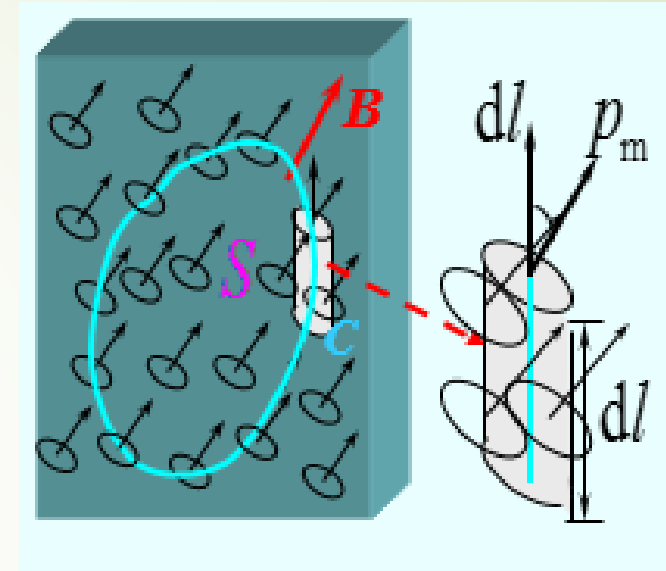
Calculation of volume magnetic current density

Calculation Principle : $I_M|_S = I_M|_C$

$$\begin{aligned} \text{As, } I_M|_{dC} &= iN_{dC} = indV = ni\Delta S dl \cos \theta \\ &= np_m dl \cos \theta = M dl \cos \theta = \vec{M} \cdot \vec{dl} \end{aligned}$$

so,
$$I_M = \oint_C \vec{M} \cdot d\vec{l}$$

→
$$\vec{J}_M = \nabla \times \vec{M}$$



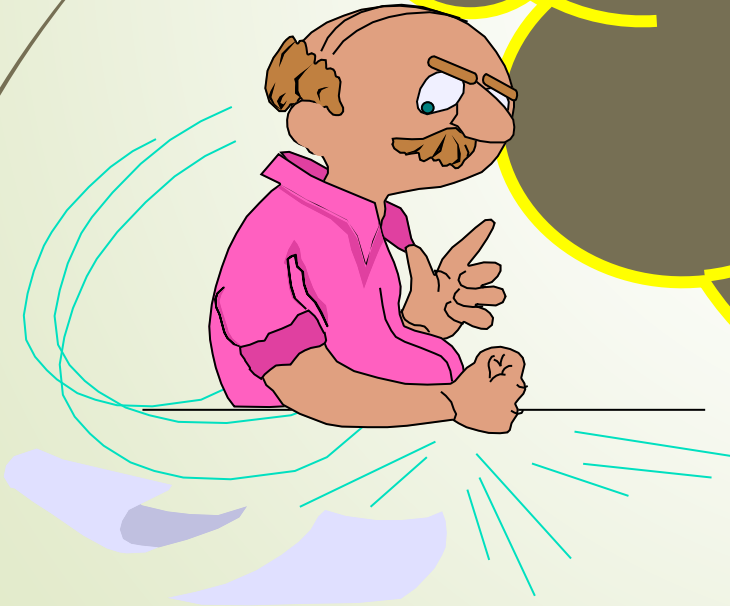
Calculation of surface magnetic current density

At the interface of media :

$$\text{As, } I_M|_{dC} = \vec{M} \cdot \vec{dl}$$

$$\text{so, } I_M|_{dC} = M_t dl \qquad J_{SM} = M_t$$

→
$$\vec{J}_{SM} = \vec{M} \times \vec{e}_n$$



Question: When magnetized current exist, **additional magnetic field will be generated** and affect the total magnetic field distribution. Then when we calculate the total field, is it necessary to calculate the magnetic field generated by the magnetic current **in advance**?

6.5 Magnetic Field Intensity and Relative Permeability

❖ The macroscopic effect of magnetization can be studied by incorporating the equivalent volume current density.

$$\oint_C \vec{B} \cdot d\vec{l} = \mu_0 \int_S (\vec{J} + \vec{J}_M) \cdot d\vec{S} \quad \longrightarrow \quad \begin{aligned} \nabla \times \vec{B} &= \mu_0 (\vec{J} + \vec{J}_M) \\ \nabla \times \vec{B} &= \mu_0 (\vec{J} + \nabla \times \vec{M}) \\ \nabla \times \left(\frac{\vec{B}}{\mu_0} - \vec{M} \right) &= \vec{J} \end{aligned}$$

Introduce magnetic field intensity:

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M}, \quad \text{so } \vec{B} = \mu_0 (\vec{H} + \vec{M})$$

then $\nabla \times \vec{H}(\vec{r}) = \vec{J}(\vec{r})$

The integral form is $\oint_C \vec{H}(\vec{r}) \cdot d\vec{l} = \int_S \vec{J}(\vec{r}) \cdot d\vec{S}$



Conclusion

When there are medium in space , the problems of **steady magnetic field** can be described by the following **basic equations**.

$$\begin{cases} \nabla \times \vec{H}(\vec{r}) = \vec{J}(\vec{r}) \\ \nabla \cdot \vec{B}(\vec{r}) = 0 \end{cases}$$

(Differential form)

$$\begin{cases} \oint_C \vec{H}(\vec{r}) \cdot d\vec{l} = \int_S \vec{J}(\vec{r}) \cdot d\vec{S} \\ \oint_S \vec{B}(\vec{r}) \cdot d\vec{S} = 0 \end{cases}$$

(integral form)

The process of solving the problems can use following route :

$$\vec{J} \longrightarrow \vec{H} \longrightarrow \vec{B}, \vec{M}, \vec{J}_M$$

Ex. There is an infinite conducting magnetic cylinder, with permeability of μ and radius of a , and a line current I exist on the axis of it. Outside the cylinder is air (μ_0), find the distribution of B , H and M outside the cylinder.

Solution: Applying Ampere's law, then

Magnetic
field intensity

$$\oint_C \vec{H} \cdot d\vec{l} = 2\pi\rho H_\phi = I$$

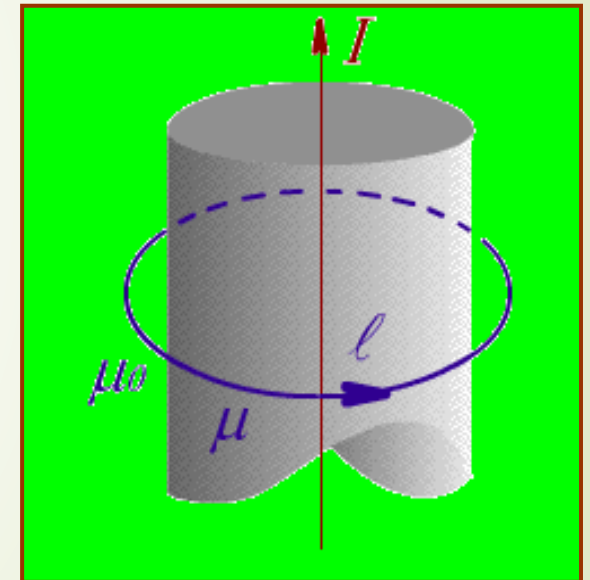
$$\vec{H} = \vec{e}_\phi \frac{I}{2\pi\rho} \quad 0 < \rho < \infty$$

Magnetic
flux density

$$\vec{B} = \begin{cases} \vec{e}_\phi \frac{\mu I}{2\pi\rho} & 0 < \rho < a \\ \vec{e}_\phi \frac{\mu_0 I}{2\pi\rho} & a < \rho < \infty \end{cases}$$

Magnetization
intensity

$$\vec{M} = \frac{\vec{B}}{\mu_0} - \vec{H} = \begin{cases} \vec{e}_\phi \frac{\mu - \mu_0}{\mu_0} \cdot \frac{I}{2\pi\rho} & \rho < a \\ 0 & a < \rho < \infty \end{cases}$$



• Classification and constitutive relation of magnetic medium

Classification :

$$\mu_r < 1$$

Diamagnetic
substance

Water: 0.99999

$$\mu_r > 1$$

Paramagnetic
substance

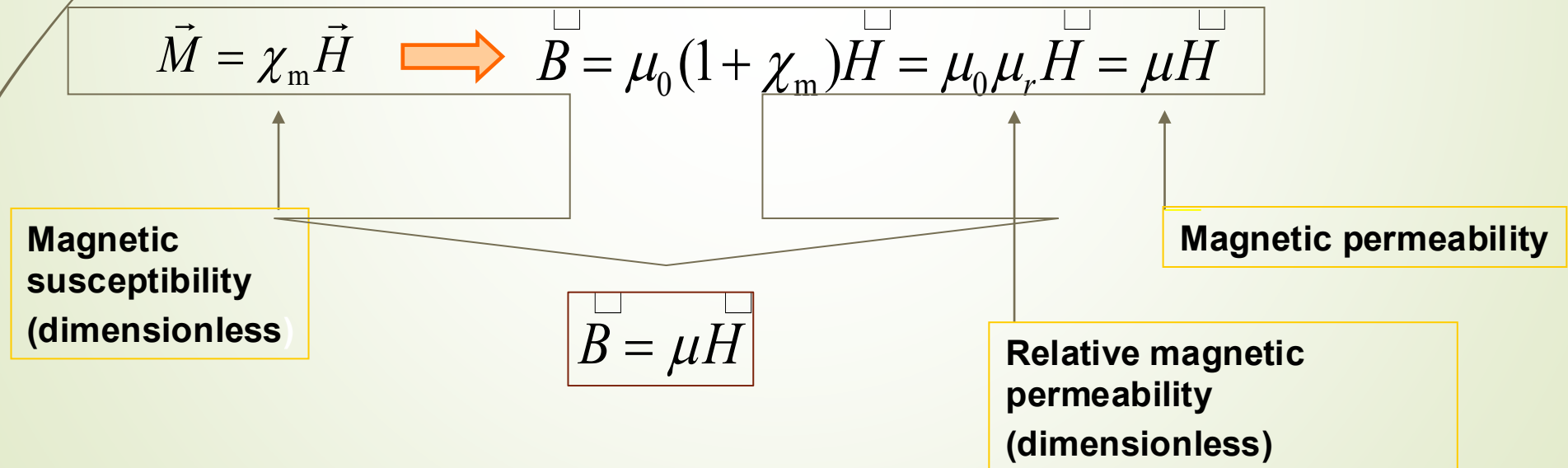
Air: 1.0000004

$$\mu_r \gg 1$$

Ferromagnetic
substance

Iron: 4000

Constitutive relation :





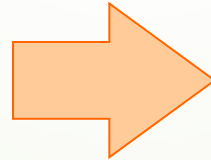
6.6 Boundary Condition for Magnetostatic Fields

1. Magnetic flux density B

Choose a point P on the interface, make a cylindrical surface S enclosed the point P

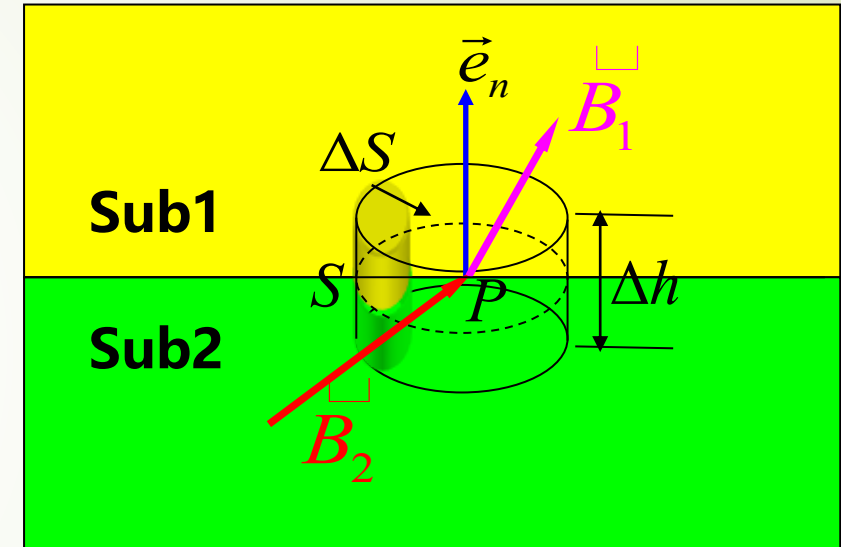
Since $\Delta h \rightarrow 0$, then

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$



$$(\vec{B}_1 - \vec{B}_2) \cdot \vec{e}_n \Delta S = 0$$

So $\vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0$ Or $B_{1n} = B_{2n}$



2. Magnetic field intensity H

Choose the closed path as the contour C at the both sides of the interface, Since $\Delta h \rightarrow 0$, then

$$\oint_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S} \quad \Rightarrow \quad (\vec{H}_1 - \vec{H}_2) \cdot \Delta \vec{l} = \vec{J}_s \cdot \vec{N} \Delta l$$

$$\Delta \vec{l} = \vec{N} \times \vec{e}_n \Delta l \quad \Rightarrow \quad (\vec{H}_1 - \vec{H}_2) \cdot \Delta \vec{l} = (\vec{H}_1 - \vec{H}_2) \cdot (\vec{N} \times \vec{e}_n) \Delta l$$

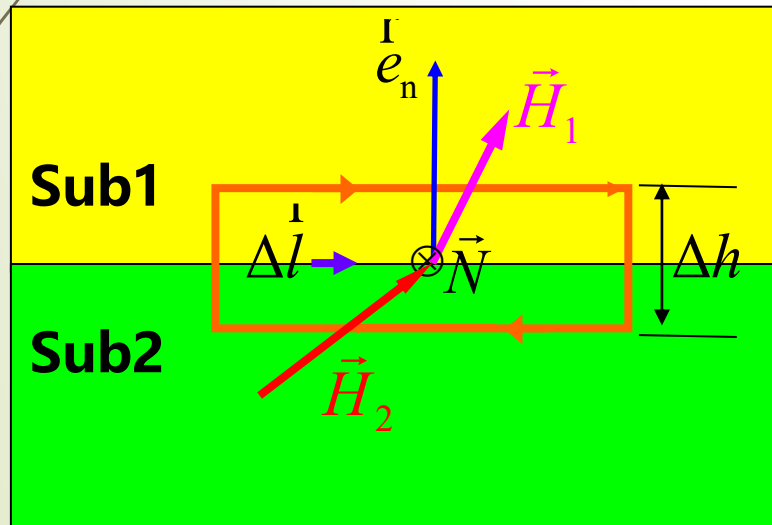
$$= [\vec{e}_n \times (\vec{H}_1 - \vec{H}_2)] \cdot \vec{N} \Delta l$$

So

$$\vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$$

Or

$$H_{1t} - H_{2t} = J_s$$



➤ Two perfect dielectric

$$\begin{cases} \mathbf{e}_n \cdot (\mathbf{D}_1 - \mathbf{D}_2) = 0 \\ \mathbf{e}_n \cdot (\mathbf{B}_1 - \mathbf{B}_2) = 0 \\ \mathbf{e}_n \times (\mathbf{E}_1 - \mathbf{E}_2) = 0 \\ \mathbf{e}_n \times (\mathbf{H}_1 - \mathbf{H}_2) = 0 \end{cases}$$

❖ Perfect dielectric and perfect conductor

$$\begin{cases} \mathbf{e}_n \cdot \mathbf{D} = \rho_s \\ \mathbf{e}_n \cdot \mathbf{B} = 0 \\ \mathbf{e}_n \times \mathbf{E} = 0 \\ \mathbf{e}_n \times \mathbf{H} = \mathbf{J}_s \end{cases}$$



➤ Boundary conditions of Magnetic vector potential

$$\vec{A}_1 = \vec{A}_2$$

$$\vec{e}_n \times \left(\frac{1}{\mu_1} \nabla \times \vec{A}_1 - \frac{1}{\mu_2} \nabla \times \vec{A}_2 \right) = \vec{J}_s$$

$$\nabla^2 \vec{A} = -\mu \vec{J}$$

$$\oint_C \vec{A} \cdot d\vec{l} = \int_S \vec{B} \cdot d\vec{S}$$

$$\Rightarrow A_{1t} = A_{2t}$$

$$\Rightarrow A_{1n} = A_{2n}$$

$$\Rightarrow \vec{A}_1 = \vec{A}_2$$

$$\nabla \cdot \vec{A} = 0$$

$$\vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s$$

$$\vec{H} = \nabla \times \vec{A} / \mu$$

$$\Rightarrow \vec{e}_n \times \left(\frac{1}{\mu_1} \nabla \times \vec{A}_1 - \frac{1}{\mu_2} \nabla \times \vec{A}_2 \right) = \vec{J}_s$$

Ex. The air between two conductor plate ($z = 0$ and $z = d$), the electric field intensity is known as

$$\vec{E} = \hat{e}_y E_0 \sin\left(\frac{\pi}{d} z\right) \cos(\omega t - k_x x) \quad \text{V/m}$$

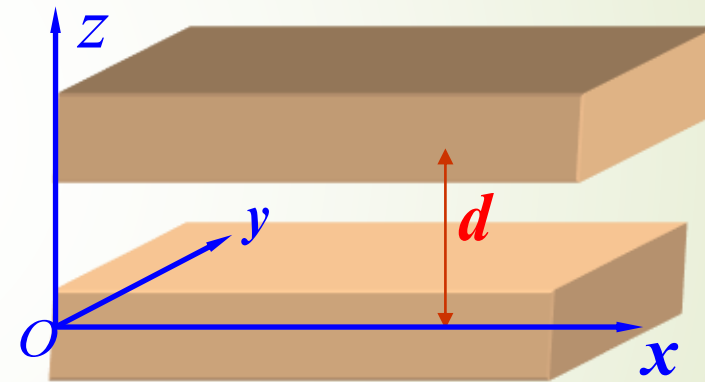
Find: (1) Magnetic field intensity $\vec{H}$;
 (2) The surface current density $\vec{J}_s$.

Solution (1) As $\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t}$

$$\frac{\partial \vec{H}}{\partial t} = -\frac{1}{\mu_0} \nabla \times \vec{E}$$

$$= -\frac{1}{\mu_0} \left(-\hat{e}_x \frac{\partial E_y}{\partial z} + \hat{e}_z \frac{\partial E_y}{\partial x} \right)$$

$$= \frac{E_0}{\mu_0} \left[\hat{e}_x \frac{\pi}{d} \cos\left(\frac{\pi}{d} z\right) \cos(\omega t - k_x x) - \hat{e}_z k_x \sin\left(\frac{\pi}{d} z\right) \sin(\omega t - k_x x) \right]$$



Calculate the time integral of above equation

$$\begin{aligned}\vec{H}(x, z, t) &= \int \frac{\partial \vec{H}(x, z, t)}{\partial t} dt \\ &= \vec{e}_x \frac{\pi E_0}{\omega \mu_0 d} \cos\left(\frac{\pi}{d} z\right) \sin(\omega t - k_x x) + \\ &\quad \vec{e}_z \frac{k_x E_0}{\omega \mu_0} \sin\left(\frac{\pi}{d} z\right) \cos(\omega t - k_x x) \quad (\text{A/m})\end{aligned}$$

(2) The current density of the conductor surface at $z = 0$

$$\vec{J}_S = \vec{e}_z \times \vec{H}\Big|_{z=0} = \vec{e}_y \frac{\pi E_0}{\omega \mu_0 d} \sin(\omega t - k_x x) \quad (\text{A/m})$$

The current density of the conductor surface at $z = d$

$$\vec{J}_S = (-\vec{e}_z) \times \vec{H}\Big|_{z=d} = \vec{e}_y \frac{\pi E_0}{\omega \mu_0 d} \sin(\omega t - k_x x) \quad (\text{A/m})$$

6.7 Inductances and Inductors

1. Magnetic Flux and Flux linkage

- Magnetic Flux (磁通)

$$\Phi = \int_S \vec{B} \cdot d\vec{S} = \int_S \nabla \times \vec{A} \cdot d\vec{S} = \oint_C \vec{A} \cdot d\vec{l}$$

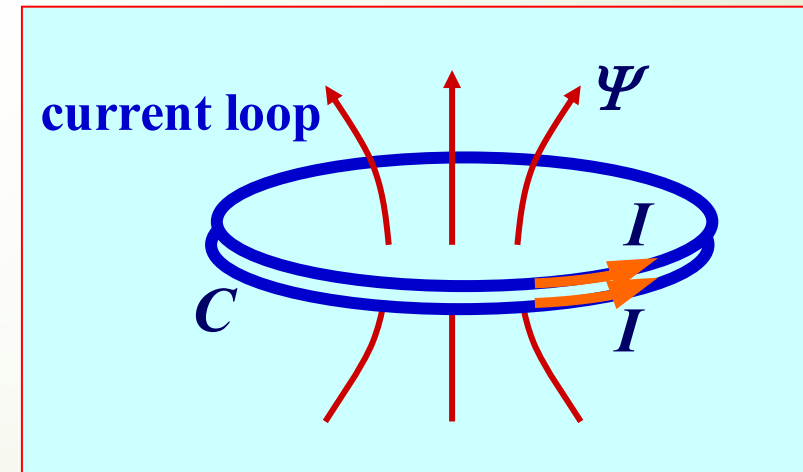
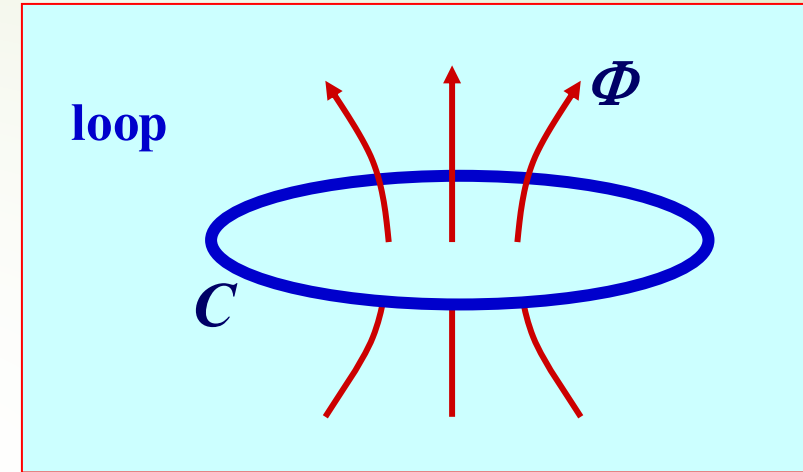
Features: the circuit can be any geometric loops

- Flux linkage (磁链)

Total flux of all current loop hinges

Features:

- The loop is current loop
- Include all the current loops
- Each loop includes the total magnetic flux linkage of the hinge



● Magnetic flux calculation

✓ Single turn coil

$$\Psi = \Phi$$

✓ Multi turn coil

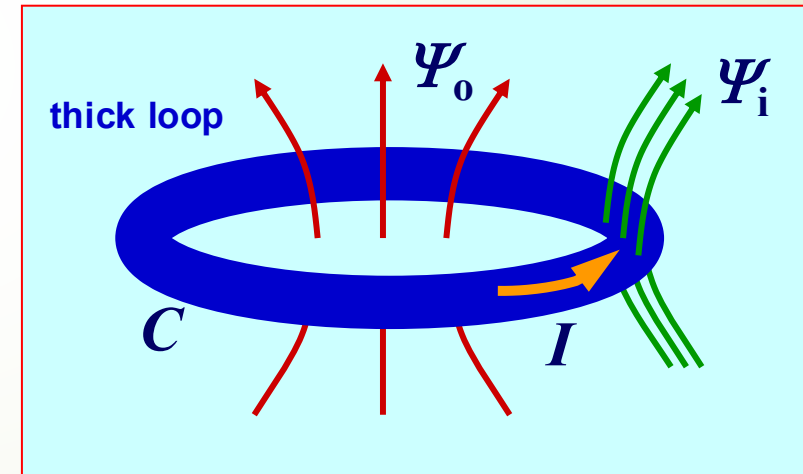
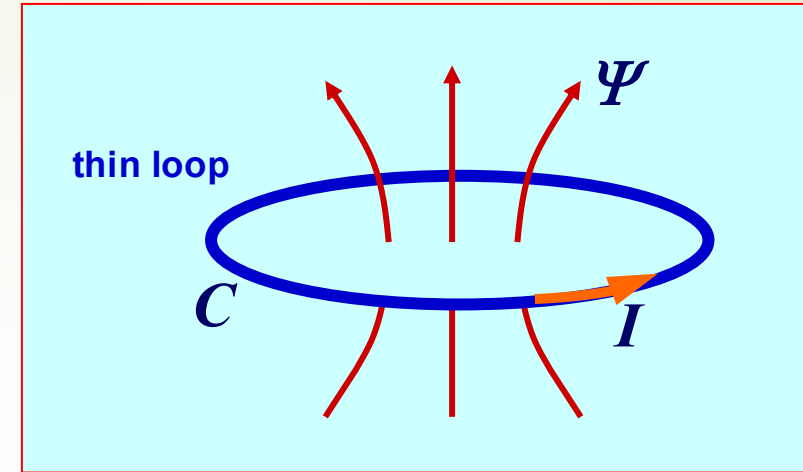
$$\Psi = n\Phi$$

n : is the ratio the current of the magnetic field hinge to the current of the circuit current I (Not necessarily integer)

✓ Thick wire loops

$$\Psi = \Psi_o + \Psi_i$$

Ψ_o : External magnetic flux ; Ψ_i : Internal magnetic flux



2. Self-inductance

$$L = \frac{\Psi}{I}$$

called the self inductance coefficient of circuit C, short for self-inductance .

Features: the magnetic flux is generated by I.

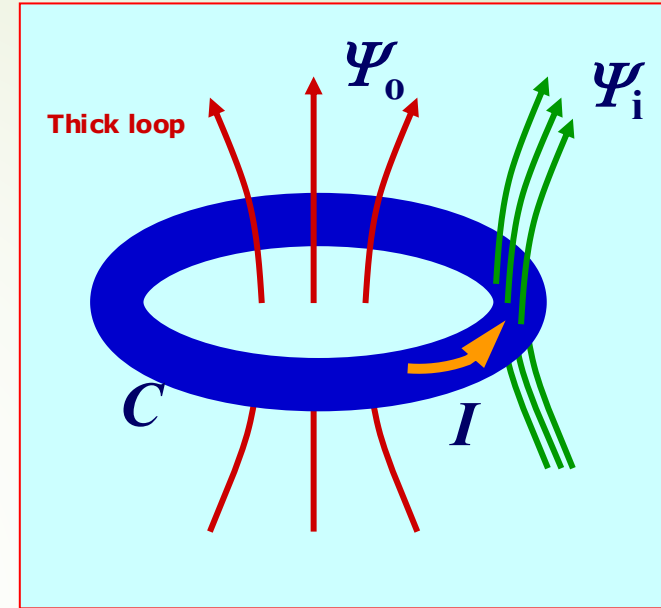
Self-inductance of thick conductor loop :

$$L = L_i + L_o$$

$$L_i = \frac{\Psi_i}{I} \quad \text{—— Internal self inductance ;} \quad L_o = \frac{\Psi_o}{I} \quad \text{—— External self inductance ;}$$

■ **The characteristics of self-inductance:**

self-inductance is related to geometry of loops , size and the surrounding magnetic medium, and not related the value of the magnetic flux and current .



3. Mutual inductance

$$M_{21} = \frac{\Psi_{21}}{I_1} \quad (M_{11} = L_1)$$

Mutual inductance coefficient of loop C_1 to loop C_2 , short for mutual inductance.

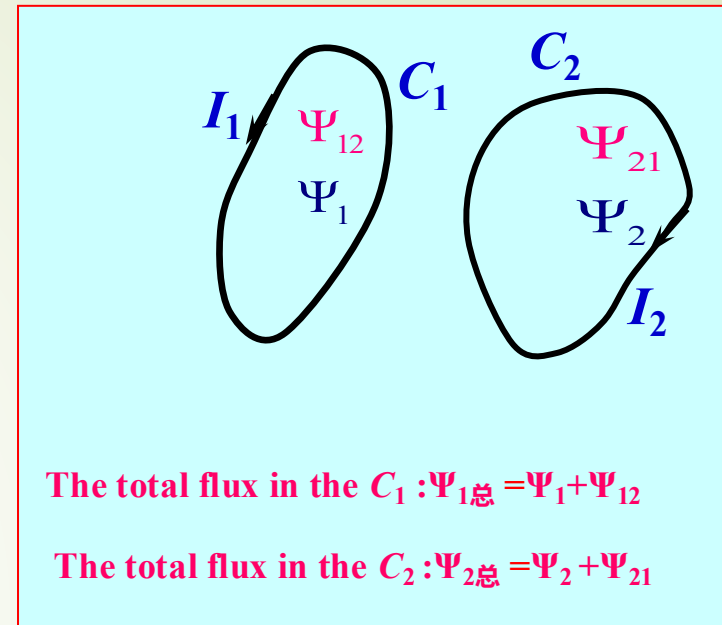
Features: the magnetic flux in the C_2 is generated by I_1

Similarly, the mutual inductance of the loop C_2 to the loop C_1 is

$$M_{12} = \frac{\Psi_{12}}{I_2}$$

Features: the magnetic flux in the C_1 is generated by I_2

Question : $\Psi_{1\text{total}} = ?$; $\Psi_{2\text{total}} = ?$



$$M_{21} = M_{12} = M$$

$$M = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{R}$$

**Neumann's
formula**

4. Proof of the Newman's formula

$$\Psi_{21} = \oint_{C_2} \vec{A}_1 \cdot d\vec{l}_2$$

And: $\vec{A}_1(\vec{r}_2) = \frac{\mu_0 I_1}{4\pi} \oint_{C_1} \frac{d\vec{l}_1}{R}$

Then:

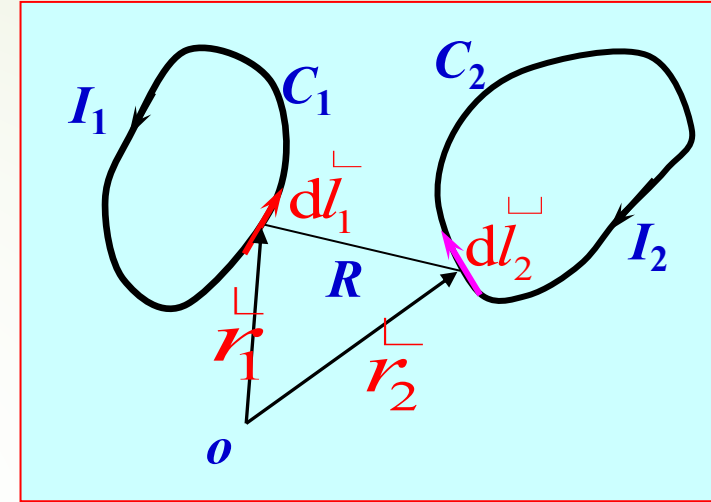
$$\Psi_{21} = \oint_{C_2} \vec{A}_1 \cdot d\vec{l}_2 = \frac{\mu_0 I_1}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\vec{l}_1 \cdot d\vec{l}_2}{R}$$

**Newman's
formula**

So: $M_{21} = \frac{\Psi_{21}}{I_1} = \frac{\mu_0}{4\pi} \oint_{C_2} \oint_{C_1} \frac{d\vec{l}_1 \cdot d\vec{l}_2}{R}$

Similarly: $M_{12} = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\vec{l}_2 \cdot d\vec{l}_1}{R} = M_{21}$

$$M = \frac{\mu_0}{4\pi} \oint_{C_1} \oint_{C_2} \frac{d\vec{l}_1 \cdot d\vec{l}_2}{R}$$



Ex. Find the self-inductance for coaxial line.

Solution: find the internal self-inductance first

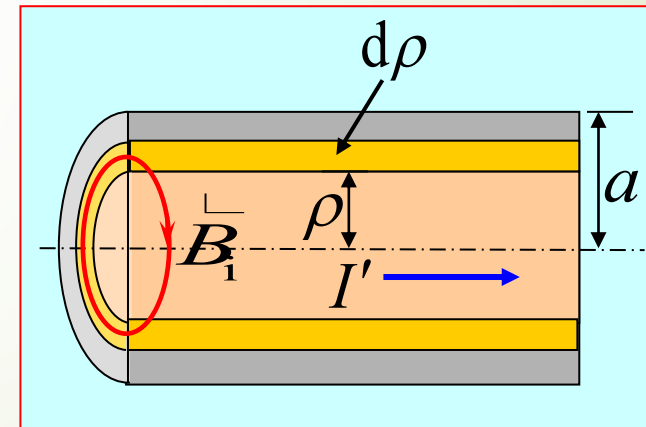
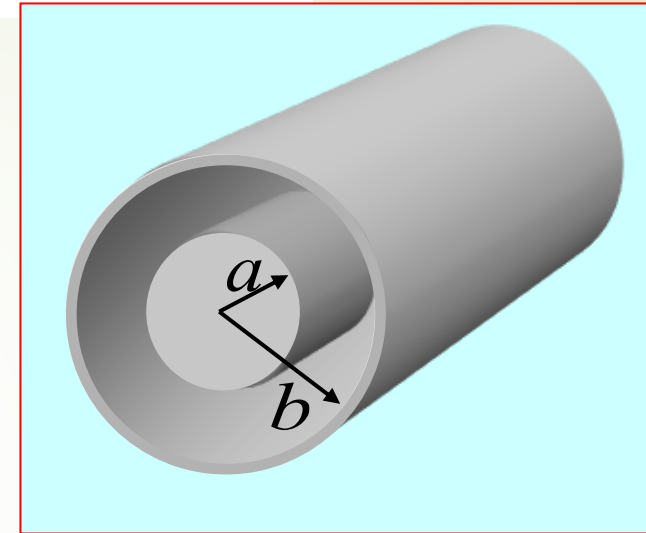
$$\oint_C \vec{H}_i \cdot d\vec{l} = I' = \frac{I}{\pi a^2} \pi \rho^2 = \frac{I}{a^2} \rho^2$$

so
$$\vec{H}_i = \frac{I}{2\pi a^2} \rho, \quad \vec{B}_i = \frac{\mu_0 I}{2\pi a^2} \rho$$

Magnetic Flux for surface element at ρ

$$d\Phi_i = \vec{B}_i \cdot d\vec{S} = \frac{\mu_0 I}{2\pi a^2} \rho l d\rho$$

Flux linkage:
$$d\Psi_i = \frac{I'}{I} d\Phi_i = \frac{\mu_0 I \rho^3}{2\pi a^4} l d\rho$$



Total flux linkage in inner conductor is

$$\Psi_i = \int d\Psi_i = l \int_0^a \frac{\mu_0 I \rho^3}{2\pi a^4} d\rho = \frac{\mu_0 I}{8\pi} l$$

**internal self-inductance
per unit length**

$$L_i = \frac{\Psi_i}{Il} = \frac{\mu_0}{8\pi}$$

Find the external self-inductance between two conductors

$$B = \frac{\mu_0 I}{2\pi\rho} \quad \longrightarrow \quad d\Psi_o = d\Phi_o = \frac{\mu_0 I}{2\pi\rho} l d\rho$$

so

$$\Psi_o = \int d\Psi_o = l \int_a^b \frac{\mu_0 I}{2\pi\rho} d\rho = l \frac{\mu_0 I}{2\pi} \ln \frac{b}{a}$$

**External self-inductance
per unit length**

$$L_o = \frac{\Psi_o}{Il} = \frac{\mu_0}{2\pi} \ln \frac{b}{a}$$

**Total self-inductance
per unit length**

$$L = L_i + L_o = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \frac{b}{a}$$



Ex. Find the self-inductance per unit length of a two-wire transmission line. ($D \gg a$)

Solution: Applying Ampere's circuital law and superposition

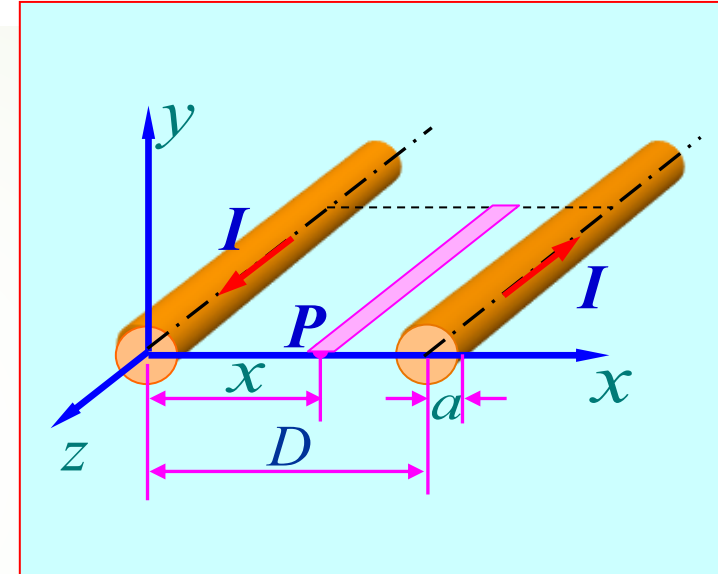
$$\vec{B}(x) = \vec{e}_y \frac{\mu_0 I}{2\pi} \left(\frac{1}{x} + \frac{1}{D-x} \right)$$

External flux linkage,

$$\Psi_o = \int \vec{B} \cdot d\vec{S} = \frac{\mu_0 I}{2\pi} l \int_a^{D-a} \left(\frac{1}{x} + \frac{1}{D-x} \right) dx = \frac{\mu_0 I}{\pi} l \ln \frac{D-a}{a}$$

External self-inductance per unit length:

$$L_o = \frac{\Psi_o}{I} = \frac{\mu_0}{\pi} \ln \frac{D-a}{a} \approx \frac{\mu_0}{\pi} \ln \frac{D}{a}$$



internal self-inductance per unit length for two wires

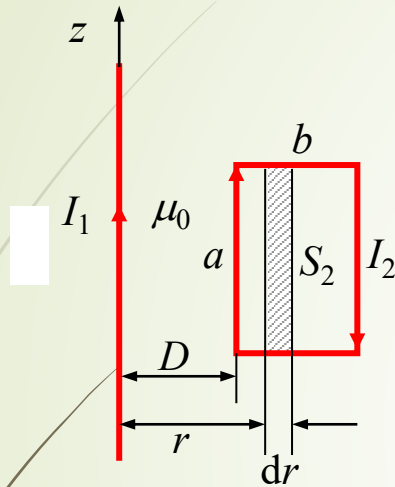
$$L_i = 2 \times \frac{\mu_0}{8\pi} = \frac{\mu_0}{4\pi}$$

The total self-inductance per unit length of a two-wire transmission line

$$L = L_i + L_o = \frac{\mu_0}{4\pi} + \frac{\mu_0}{\pi} \ln \frac{D}{a}$$



Ex. Calculate the mutual inductance between an infinitely long straight line and a rectangular coil. The line and the coil are at the same plane, and in vacuum.



Solution: Select cylindrical coordinate system, and let the infinitely long straight line to be at the z -axis. The magnetic flux density produced by current I_1 is then

$$\mathbf{B}_1 = \frac{\mu_0 I_1}{2\pi r} \mathbf{e}_\phi$$

The magnetic flux linkage Ψ_{21} with current I_2 by current I_1 is

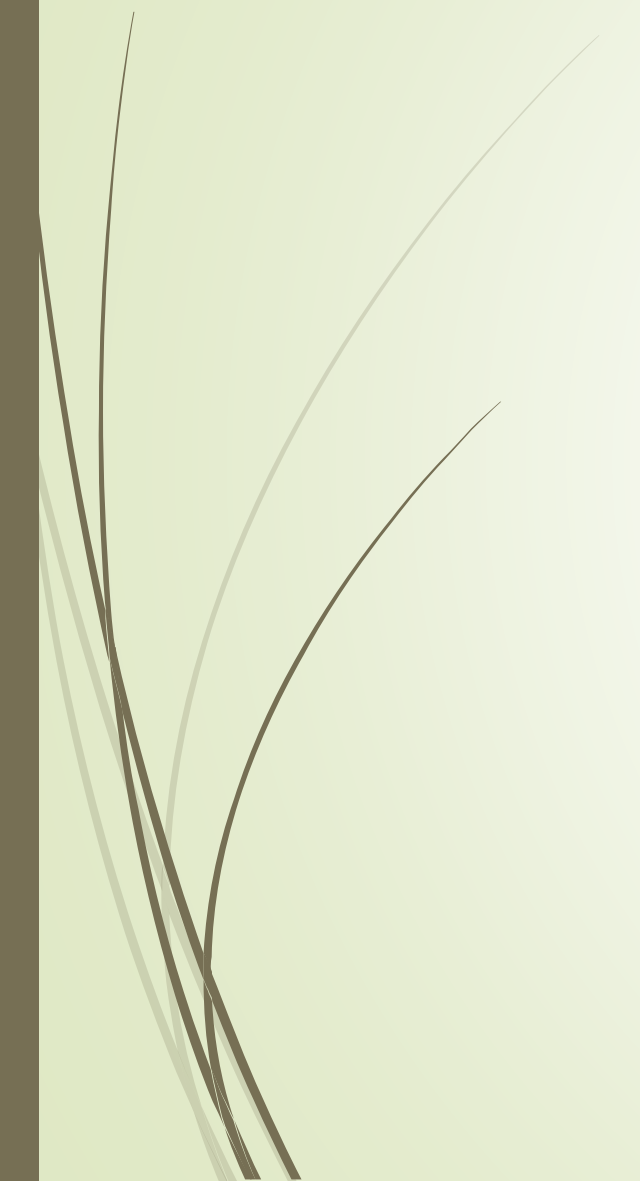
$$\Psi_{21} = \int_{S_2} \mathbf{B}_1 \cdot d\mathbf{S}$$

If the flowing direction of the current I_2 is clockwise, $d\mathbf{S}$ and $\mathbf{B}_1$ have the same direction. Then

$$\Psi_{21} = \frac{\mu_0 I_1 a}{2\pi} \int_D^{D+b} \frac{1}{r} dr = \frac{\mu_0 I_1 a}{2\pi} \ln\left(\frac{D+b}{D}\right)$$

We have
$$M_{21} = \frac{\Psi_{21}}{I_1} = \frac{\mu_0 a}{2\pi} \ln\left(\frac{D+b}{D}\right) > 0$$

If the flowing direction of the current I_2 is **counter** clockwise, then the $\mathbf{B}_1$ and $d\mathbf{S}$ are opposite, and $M_{21} < 0$.



https://www.bilibili.com/video/BV1t64y1C7o7/?spm_id_from=333.337.search-card.all.click&vd_source=cbc26ac5a27dd541735237c7e8b5a6a4

6.8 Magnetic Energy

- Experimental evidence of magnetic field energy

$$\vec{F} = \oint_c \vec{N} d\vec{l} \times \vec{B}$$

- Distribution space of magnetic field energy

Where there is a magnetic field, there is the energy of the magnetic field !



1. Calculation of magnetic field energy by magnetic field distribution

- Energy density of magnetic field :

$$w_m = \frac{1}{2} \vec{B} \cdot \vec{H}$$

- Energy of magnetic field :

$$W_m = \frac{1}{2} \int_V \vec{B} \cdot \vec{H} dV$$

The magnetic field energy
in the integral domain V

- For linear, isotropic medium it can be written as:

$$w_m = \frac{1}{2} \vec{B} \cdot \vec{H} = \frac{1}{2} \mu \vec{H} \cdot \vec{H} = \frac{1}{2} \mu H^2$$

$$W_m = \frac{1}{2} \int_V \vec{B} \cdot \vec{H} dV = \frac{1}{2} \int_V \mu \vec{H} \cdot \vec{H} dV = \frac{1}{2} \int_V \mu H^2 dV$$



2. Calculation of magnetic field energy by magnetic vector potential

✓ Volume current distribution

$$W_m = \frac{1}{2} \int_V \mathbf{J} \cdot \mathbf{A} dV$$

✓ Surface current distribution

$$W_m = \frac{1}{2} \int_S \mathbf{J}_s \cdot \mathbf{A} dS$$

✓ Line current of a circuit loop

$$W_m = \frac{I}{2} \oint_C \mathbf{A} \cdot d\mathbf{l} = \frac{I}{2} \psi_{\text{total}}$$



3. Calculation of magnetic field energy by inductance

● Inductive energy storage
(Single current-carrying loop) :

$$W_m = \frac{I}{2} \psi_{\text{总}} = \frac{1}{2} I^2 L$$

● Inductive energy storage
(N current-carrying loops) :

$$\begin{aligned} W_m &= \frac{1}{2} \sum_{j=1}^N I_j \Psi_{j\text{总}} = \frac{1}{2} \sum_{j=1}^N I_j \sum_k^N \Psi_{jk} \\ &= \frac{1}{2} \sum_{j=1}^N \sum_k^N I_j I_k M_{jk} \end{aligned}$$

For example, two loops current-carrying

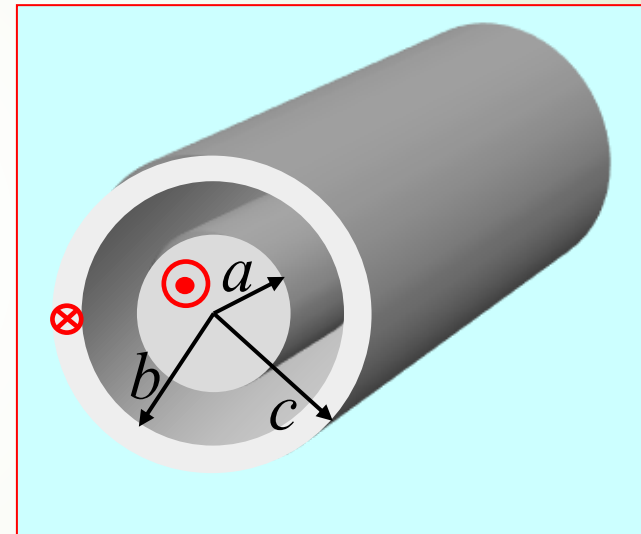
$$W_m = \frac{1}{2} \sum_{j=1}^N \sum_k^N I_j I_k M_{jk} = \frac{1}{2} L_1 I_1^2 + \frac{1}{2} L_2 I_2^2 + M I_1 I_2$$



Ex. Find the magnetic energy and self inductance in a coaxial line per unit length.

Solution: from Ampere's circuital law

$$\vec{H} = \begin{cases} \hat{e}_\phi \frac{\rho I}{2\pi a^2} & 0 < \rho < a \\ \hat{e}_\phi \frac{I}{2\pi\rho} & a < \rho < b \\ \hat{e}_\phi \frac{I}{2\pi\rho} \frac{c^2 - \rho^2}{c^2 - b^2} & b < \rho < c \\ 0 & \rho > c \end{cases}$$



The magnetic energy per unit length in three area:

$$W_{m1} = \frac{\mu_0}{2} \int_0^a \left(\frac{\rho I}{2\pi a^2} \right)^2 2\pi \rho d\rho = \frac{\mu_0 I^2}{16\pi}$$

$$W_{m2} = \frac{\mu_0}{2} \int_a^b \left(\frac{I}{2\pi \rho} \right)^2 2\pi \rho d\rho = \frac{\mu_0 I^2}{4\pi} \ln \frac{b}{a}$$

$$\begin{aligned} W_{m3} &= \frac{\mu_0}{2} \int_b^c \left(\frac{I}{2\pi \rho} \right)^2 \left(\frac{c^2 - \rho^2}{c^2 - b^2} \right)^2 2\pi \rho d\rho \\ &= \frac{\mu_0 I^2}{4\pi} \left[\frac{c^4}{(c^2 - b^2)^2} \ln \frac{c}{b} - \frac{3c^2 - b^2}{4(c^2 - b^2)} \right] \end{aligned}$$

The total magnetic energy per unit length

$$\begin{aligned} W_m &= W_{m1} + W_{m2} + W_{m3} \\ &= \frac{\mu_0 I^2}{16\pi} + \frac{\mu_0 I^2}{4\pi} \ln \frac{b}{a} + \frac{\mu_0 I^2}{4\pi} \left[\frac{c^4}{(c^2 - b^2)^2} \ln \frac{c}{b} - \frac{3c^2 - b^2}{4(c^2 - b^2)} \right] \end{aligned}$$

The total self-inductance per unit length

$$L = \frac{2W_m}{I^2} = \frac{\mu_0}{8\pi} + \frac{\mu_0}{2\pi} \ln \frac{b}{a} + \frac{\mu_0}{2\pi} \left[\frac{c^4}{(c^2 - b^2)^2} \ln \frac{c}{b} - \frac{3c^2 - b^2}{4(c^2 - b^2)} \right]$$

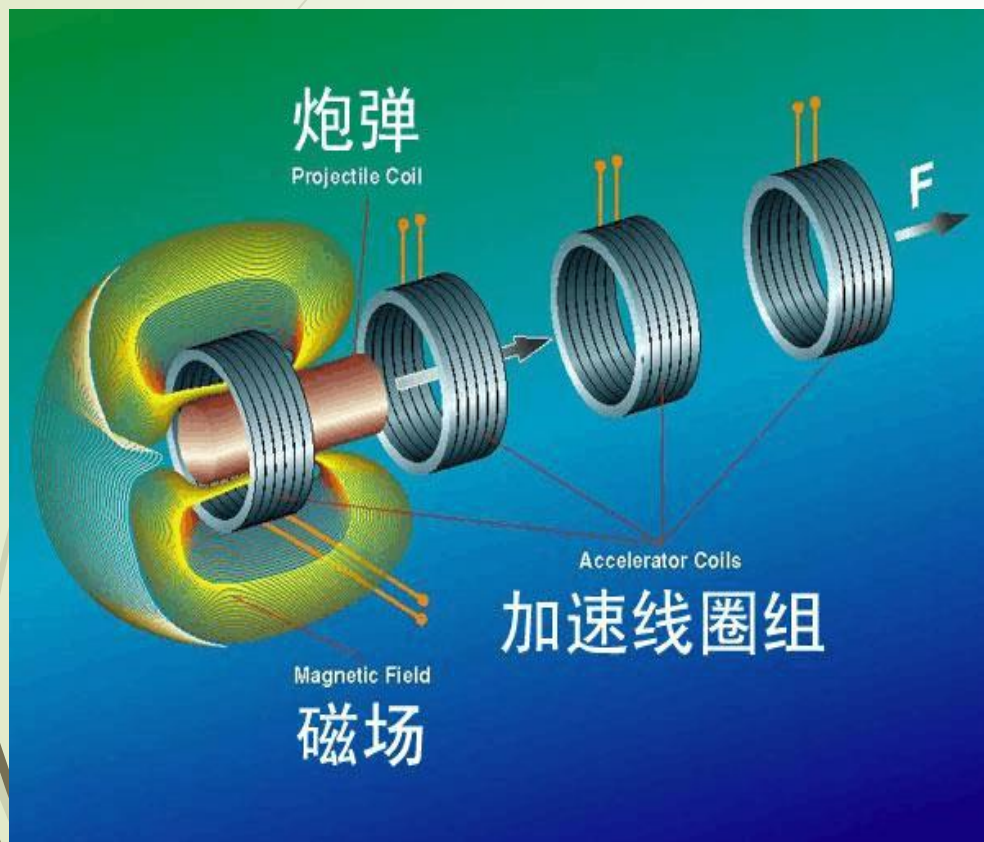
Internal self-inductance
for inner conductor

External self-inductance
between two conductors

Internal self-
inductance
for outer
conductor

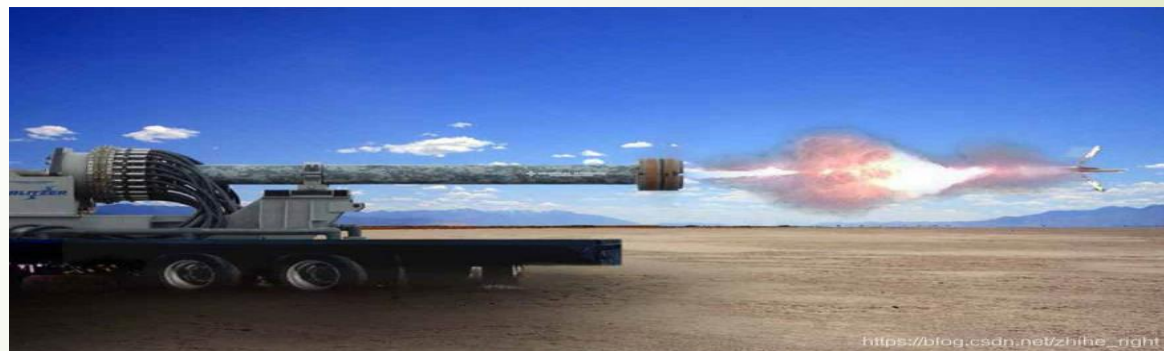
知识拓展：电磁炮

什么是电磁炮？



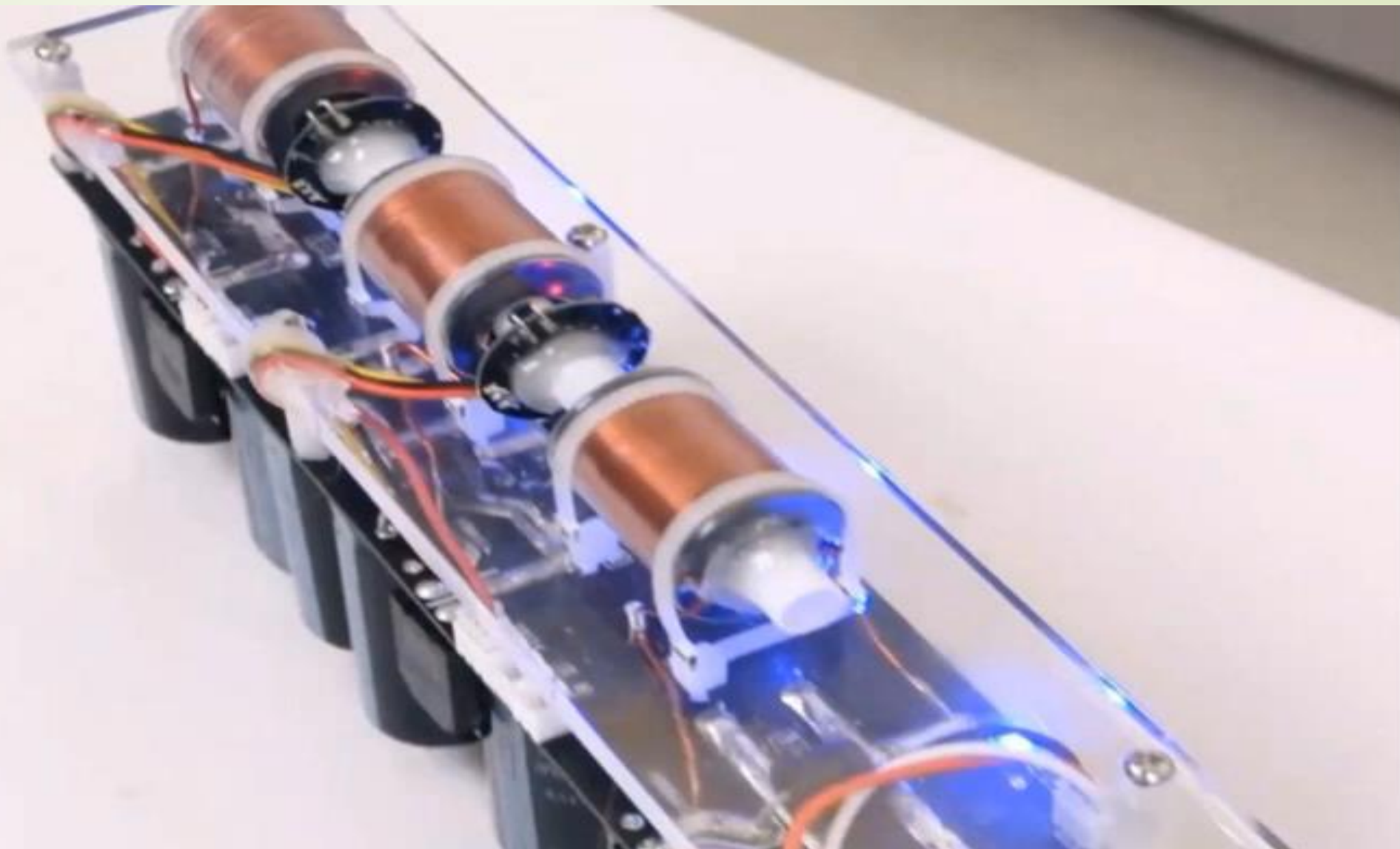
电磁炮有什么优点？

1. 弹丸初速高，射程远；
2. 结构简单，没有弹壳、药筒
3. 安全可靠，无炸药不会出现储存不当的爆炸。
4. 成本低：仅需电力。

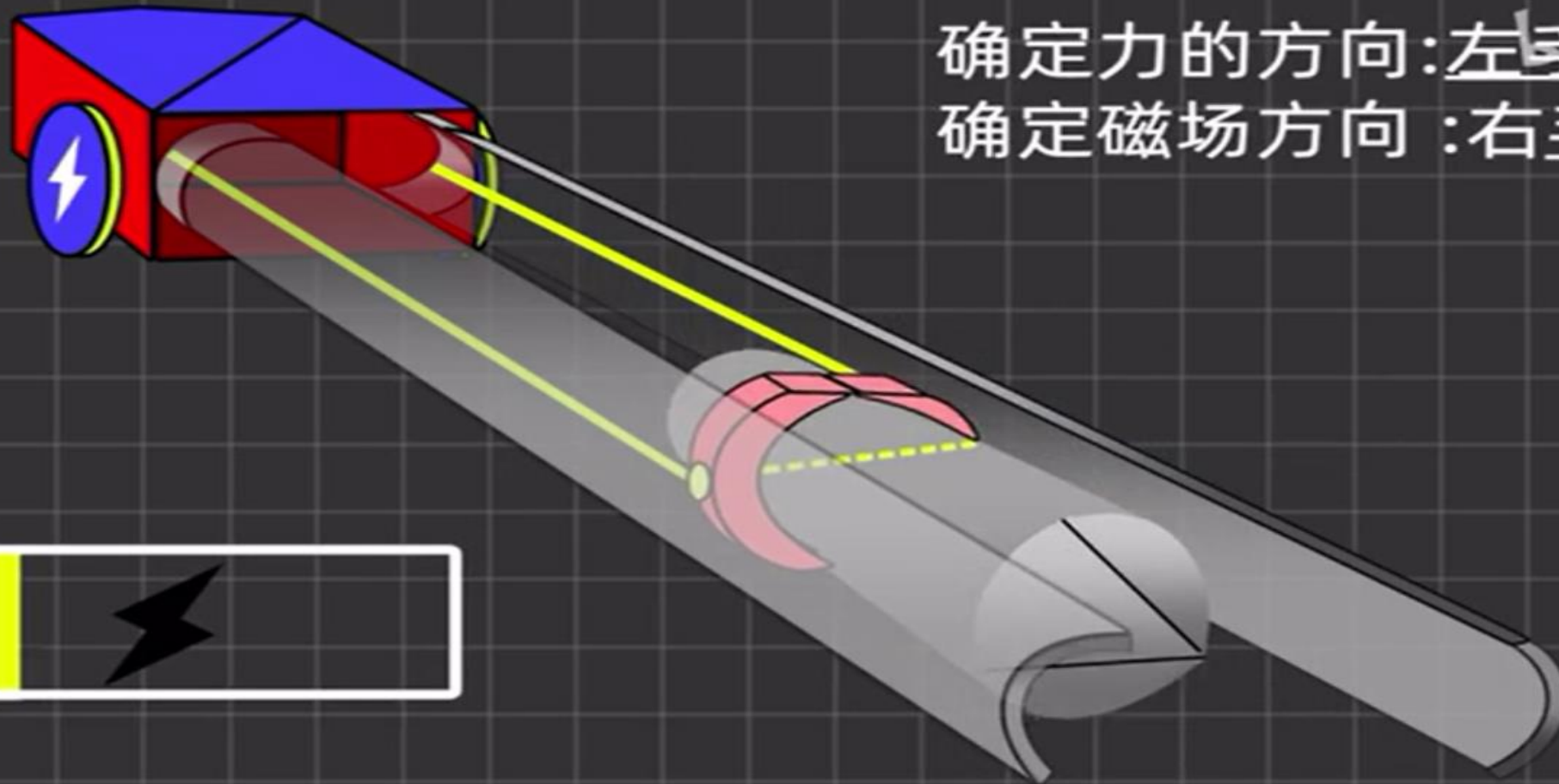


静电场的应用：电磁线圈炮（线圈间相互作用力）

@别小齐



静电场的应用：电磁轨道炮（安培力）



确定力的方向：左手定则
确定磁场方向：右手定则

只要通过的电流足够大

知识拓展：航母电磁弹射器

国别	舰种	舰级	现役数（艘）	方式	排水量（万吨）	可载战斗机数	航母战力评估
美国	重型核航母	福特级	1	电磁	10	60+	★★★★★★+
	重型核航母	尼米兹级	10	蒸汽	9.5	60+	★★★★★★
中国	重型航母	福建号	1	电磁	8+	40+	★★★★★
	中型航母	山东号	1	滑越	6	24+	★★★
	中型航母	辽宁号	1	滑越	6	24+	★★★
英国	中型航母	伊丽莎白女王级	2	滑越	6.5	40	★★★★
法国	中型核航母	戴高乐号	1	蒸汽	4.2	36	★★★+
俄罗斯	中型航母	库兹涅佐夫号	1	滑越	6	24	★
印度	中型航母	维克拉马迪亚号	1	滑越	4.5	24	★
	中型航母	维特兰特号	1	滑越	4	24	★
意大利	轻型航母	加富尔号	1	滑越	2.7	16	★
	轻型航母	加里波第号	1	滑越	1.3	12	★
合计			22				

知识拓展：航母电磁弹射器

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滑越式：山东号航空母舰

优点：简单

缺点：倾斜甲板不能放飞机
可载飞机数少



尼米兹级航空母舰

英国人搞出了蒸汽弹射式。

结构复杂，英国人弃用，美国人发扬光大。

知识拓展：航母电磁弹射器

采用蒸汽弹射的国家

法国（夏尔戴高乐号，全套美国设备）

美国（10艘航母）

优点：

相对于滑越式能多装载飞机

起飞距离短

缺点：

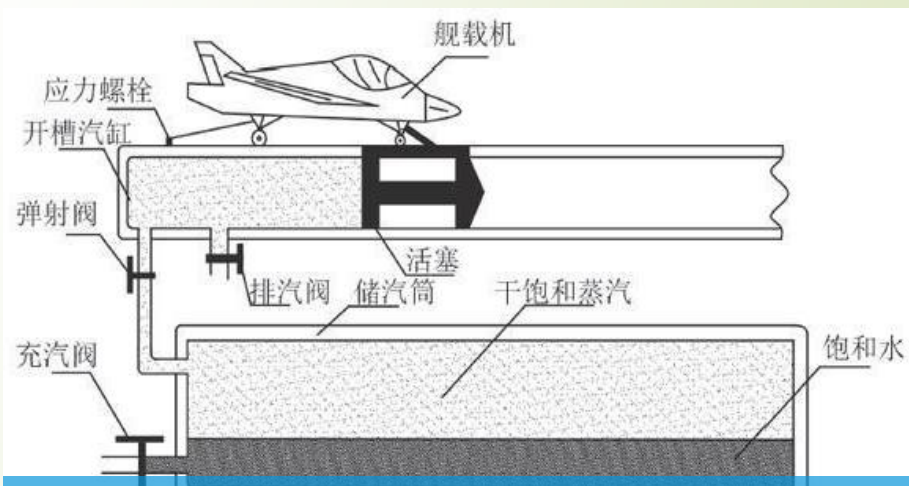
过载6G加速度

加速不均匀

无法调整弹射力度，不能适应多种机型



蒸汽管



蒸汽弹射原理

知识拓展：航母电磁弹射器

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采用电磁弹射的国家：

中国（福建号）

美国（福特级）

美国缺点：

使用交流电，不好调功率，线圈重，故障率高，发热量大，散热难。储存释放和全舰电力系统直接连通，易造成瞬间系统过载造成全舰断电。特朗普甚至扬言拆掉它，改装老版的蒸汽弹射器



福建号



福特级

知识拓展：航母电磁弹射器原理



原理与电磁轨道炮类似，实现的技术难度非常大。

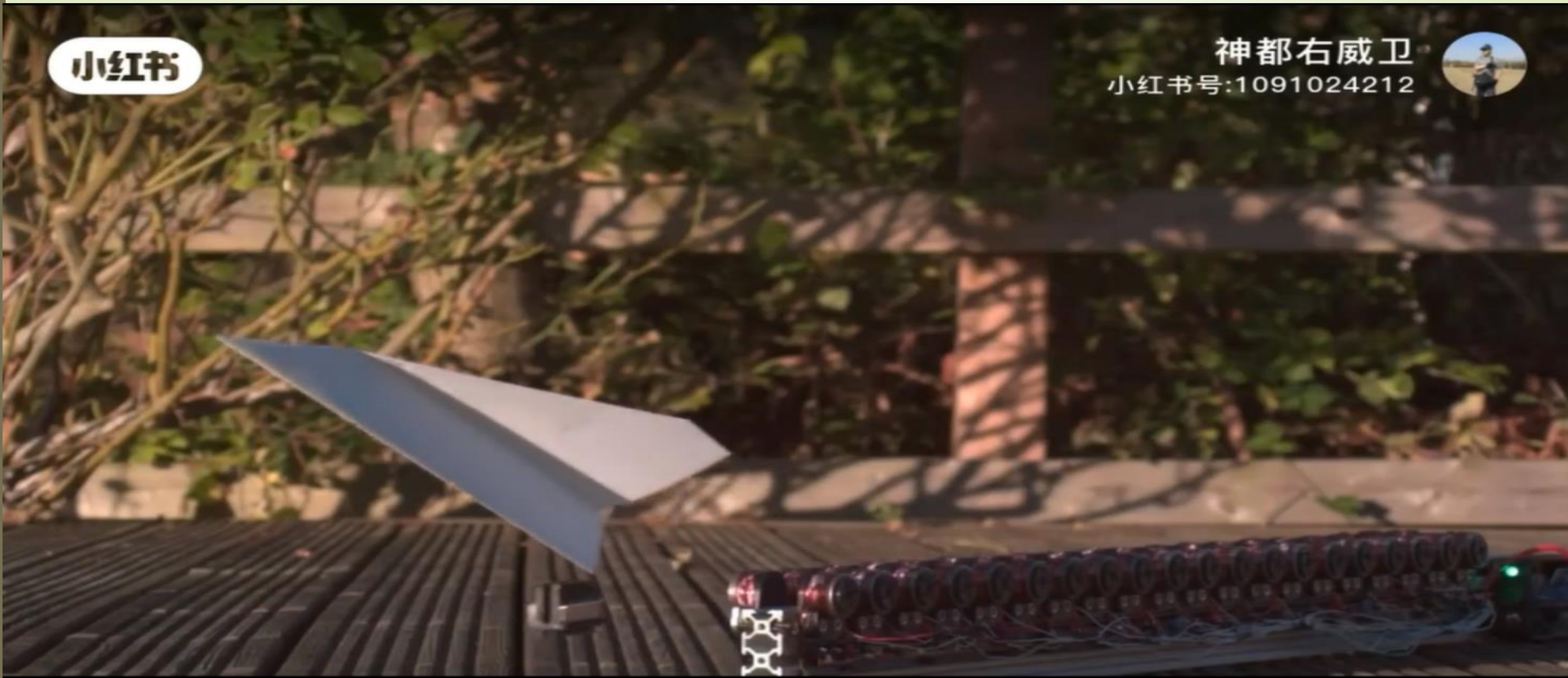
电磁线圈炮弹射出去的是几十克的子弹或者几十公斤的炮弹。

电磁弹射器弹射出去的是几十吨的飞机。

知识拓展：航母电磁弹射器原理

小红书

神都右威卫
小红书号:1091024212



Summary

1. Basic equations (general problems):

Differential forms :

$$\begin{cases} \nabla \times \vec{H} = \vec{J} \\ \nabla \cdot \vec{B} = 0 \end{cases}$$

Constitutive relations: $\vec{B} = \mu \vec{H}$

Integral forms:

$$\begin{cases} \oint_C \vec{H} \cdot d\vec{l} = \int_S \vec{J} \cdot d\vec{S} \\ \oint_S \vec{B} \cdot d\vec{S} = 0 \end{cases}$$

2. Boundary conditions (general problems):

$$\begin{cases} \vec{e}_n \cdot (\vec{B}_1 - \vec{B}_2) = 0 \\ \vec{e}_n \times (\vec{H}_1 - \vec{H}_2) = \vec{J}_s \end{cases} \quad \text{OR} \quad \begin{cases} B_{1n} - B_{2n} = 0 \\ H_{1t} - H_{2t} = J_s \end{cases}$$

3. Magnetized current density:

$$\vec{J}_M = \nabla \times \vec{M} \quad \vec{J}_{SM} = \vec{M} \times \vec{e}_n$$

4. Inductance:

$$L = \frac{\Psi}{I} \quad L = L_i + L_o$$

5. Magnetic energy density:

$$w_m = \frac{1}{2} \vec{B} \cdot \vec{H}$$

6. Magnetic energy:

$$W_m = \frac{1}{2} \int_V \vec{B} \cdot \vec{H} dV \quad W_m = \frac{1}{2} \int_V \vec{J} \cdot \vec{A} dV \quad W_m = \frac{I}{2} \psi_{\text{总}} = \frac{1}{2} I^2 L$$



Homework

According to the following link, explain the mechanism of Maglev train

https://www.bilibili.com/video/BV1sP4y1X7gX/?spm_id_from=333.337.search-card.all.click&vd_source=cbc26ac5a27dd541735237c7e8b5a6a4

 **6-4, 6-6, 6-10, 6-15, 6-22, 6-27, 6-36, 6-41**